Search for time-reversal-symmetry violation in thallium fluoride using a cryogenic buffer-gas beam source

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@Dave: Do you agree with this structure?

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The CENTREX experiment aims to provide a 30-fold increase in sensitivity over the best present upper bounds on the strength of time-violating fundamental interactions by measuring shifts in nuclear resonant frequencies in thallium fluoride. I describe the motivation for studying these time-violating interactions, and provide an overview of past work. I present the current measurement scheme for this experiment. Since the apparatus is still under construction, I present the components already built, give preliminary data collected to date, and estimate the timeline for completion of the experiment.

I. INTRODUCTION

The CENTREX experiment aims to employ a molecular-beam resonance technique to measure the nuclear Schiff moment of ^{205}Tl in thallium fluoride as a test of physics beyond the Standard Model. In the following, I outline the motivation from *CP*-violation and baryon asymmetry, define the Schiff moment and suggest how it can be measured, review relevant features of the TlF molecule, present previous experiments, the measurement scheme proposed in this work, and the projected sensitivity increase over the state of the art.

A. Motivation

Before the early 1960s, it was believed that *CP* is a good symmetry of nature. Landau pointed out that that makes it impossible for fundamental particles to have electric dipole moments [1]. When beginning their neutron EDM experiment, Purcell and Ramsey were aware of this objection [2]. Indeed, were such an EDM to be detected, that would provide clear evidence of *CP*-violation (CPV).

It has been found since that while most processes preserve *CP*, certain rare weak interactions violate it as observed in neutral *K* decays at the 0.003 level [3] and other systems, where it can be as high as 0.7. [4, chapter 13]. To account for the experimental evidence, the flavor-changing part of the standard model quark sector includes a CPV phase in the CKM quark-mixing matrix. [5] This so-called Kobayashi–Maskawa mechanism, which introduced the third quark generation precisely to explain CPV [6], agrees with all particle physics experiments to date. [7]

No corresponding flavor-neutral CPV has been observed yet. In principle, the QCD Lagrangian could contain a term of the form [8, eqn 1]

$$\mathcal{L} = \theta \frac{g^2}{32\pi^2} G^a_{\mu\nu} \widetilde{G}^a_{\mu\nu}, \tag{1}$$

where G^a is the gluon field tensor. Experimental evidence from neutral ^{199}Hg atoms [9] and neutron EDM experiments [10] suggests that the strength of this term relative to the usual strong interaction is $|\theta| < 10^{-10}$. Presently accepted theories cannot explain why θ should be that small; this fine-tuning issue is known as the strong *CP* problem. See [11] for a brief review of a few possible solutions.

At present, searches for EDMs give the most sensitive constraints on flavor-diagonal CPV. Conversely, given a particle physics model with CPV we can predict what the expected EDM of the particles is. For example, in the Standard Model, the CKM phase can produce a lepton EDM no larger than 10^{-38} e cm. Part of the reason the EDM is so suppressed is that it cannot appear below the four-loop level. [12] Given how small these effects are in the Standard Model, EDMs thus provide nearly background-free evidence of new physics. In fact, the CKM background would only become apparent when probing effects beyond the energy scale of 10^{11} GeV. [13]

The second major motivation for searching for new physics comes from the baryon asymmetry of the universe (BAU). Compared to the current number density of photons in the universe ($n_\gamma \approx 411$ cm^{-3}), the current baryon density ($n_B \approx 2.54 \cdot 10^{-7}$ cm^{-3}) is some $1.6 \cdot 10^9$ times smaller [14, Table 17]. In contrast, the antibaryon density $n_{\bar{B}}$ is smaller still; the range of upper bounds for the antimatter-to-matter number ratio has been reported as 10^{-15} to 10^{-6} [15]. I should note that the antimatter-to-matter ratio tends not to be used as a measure of BAU; instead, authors tend to prefer the ratio

$$\eta = \frac{n_B - n_{\bar{B}}}{n_\gamma} \approx \frac{n_B}{n_\gamma} \approx 10^{-10}. \tag{2}$$

There is as of yet no commonly accepted baryogenesis mechanism, i.e., explanation for how the imbalance between the number of particles and antiparticles—the C asymmetry—came to be. In a 1967 paper, Sakharov hypothesized that CPV is necessary to explain the present C asymmetry [16] if we assume the initial conditions of the universe were C-symmetric.

However, it is claimed that the 'amount' or the 'strength' of CPV in the CKM matrix is not enough to explain the extent of the current BAU. A common argument to this effect compares the size of the dimensionless constants used to parametrize the two effects. For BAU we have $\eta \approx 10^{-10}$ as defined above, whereas for CPV the constant is no larger than 10^{-20} [17, sect 2.3]. The latter number refers to the CPV invariant that can be defined in terms of quark masses as [7, eqn 3]

$$\begin{aligned} J = A(m_t^2 - m_u^2)(m_t^2 - m_c^2)(m_c^2 - m_u^2) \\ (m_b^2 - m_d^2)(m_b^2 - m_s^2)(m_s^2 - m_d^2), \end{aligned} \tag{3}$$

where A is twice the area of any "unitarity triangle" of the CKM matrix (see [4, fig 12.1]). With quark masses from [4, p40, "Quark Summary Table"] and $A \approx 3{\cdot}10^{-5}$ from [18, p7], we get $J \approx 121 \cdot 10^3$ GeV^{12}. The electroweak phase transition, occurring at temperature $T_{\mathrm{EW}} \approx 100$ GeV, was the latest point in the evolution of the universe when BAU could have been generated [19]. Comparing J and T_{EW}, we get the desired dimensionless quantity $J/T_{\mathrm{EW}}^{12} \approx 10^{-20}$ to within factors of order unity and we see it is ten orders of magnitude smaller than the constant parametrizing the baryon asymmetry. Thus, new sources of CPV are needed.

B. Schiff moment

Additional T-violating forces would likely give rise to a nuclear EDM, whether through the EDM of unpaired protons or neutrons, via CPV interactions between valence nucleons and the core of the nucleus, or other mechanisms. [20] The EDM $\mathbf{d}$ would result from a redistribution of nuclear charge density and would violate T, as follows. Since the nuclear spin $\mathbf{I}$ is the only preferred direction in a nucleus, the dipole moment also has to point along it: $\mathbf{d} = d\mathbf{I}/I$. Under time-reversal, the EDM would stay the same but the spin would flip direction.

However, according to the Schiff theorem, an EDM of a nonrelativistic point nucleus would not be observable outside of the atom. More precisely, the interaction energy of nonrelativistic point charged electric dipoles in an arbitrary external electrostatic potential has no term linear in the EDM [21]. Physically, the system rearranges itself so as to screen the external field completely [20, p33]. Thus, a T-violating EDM, if existent, would be very hard to detect experimentally without a mechanism to bypass Schiff's theorem.

In the following, we work out the shielding of a nuclear EDM in an atom with a Hamiltonian $\hat{H}$. In particular, this simple model does not attempt to take into account the complexity of molecular structure. The argument consists of the following steps: (1) Write down the atomic Hamiltonian $\hat{H}$. (2) Extract T-odd terms from the multipole expansion of $\hat{H}$. (3) Show that for a point nucleus, these terms can be written as a commutator $[\hat{H}, \ldots]$ and thus have a vanishing expectation value and no corresponding linear energy shift. (4) Relax the point-nucleus assumption, and find the residual (unscreened) interaction. To find the expectation value (or the linear energy shift) due to the interaction, two conditions need to be satisfied: the atom needs to be polarized by an external electric field (step 5), and the interaction needs to have a nonzero matrix element between states of opposite parity (step 6). Finally, we interpret $\mathbf{S}$ in terms of nuclear electric field (step 7).

Step 1: Atomic Hamiltonian Consider an atom of proton number Z in an external electric field $\mathbf{E}_0$, with electrons of momenta $\mathbf{p}_i$ at positions $\mathbf{r}_i$, and a nucleus with the charge distribution $\rho(\mathbf{R})$. The atom's Hamiltonian includes the following interactions:

$$\begin{aligned} \hat{H} \;=\; & \sum_{i=1}^{Z} \frac{\mathbf{p}_i^2}{2m} && \text{(electron kinetic energy)} \\ & + \frac{1}{2}\sum_{i,j} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_k|} && \text{(electron–electron)} \\ & - Ze^2 \sum_i \int \frac{\rho(\mathbf{R})\mathrm{d}^3\mathbf{R}}{|\mathbf{r}_i - \mathbf{R}|} && \text{(electron–nucleus)} \\ & + e\sum_i \mathbf{r}_i \cdot \mathbf{E}_0 && \text{(electrons in external field)} \\ & - Ze\langle \mathbf{R}\rangle \cdot \mathbf{E}_0. && \text{(nuclear dipole in external field)} \end{aligned}$$

The nuclear charge density is normalized to $\int \rho(\mathbf{R})\mathrm{d}^3\mathbf{R} = 1$. Almost all the integrals in this section are weighted by $\rho(\mathbf{R})$, suggesting the use of the measure $đ^3(\mathbf{R}) \equiv \rho(\mathbf{R})\mathrm{d}^3(\mathbf{R})$. Thus we have the dipole moment $\langle \mathbf{R}\rangle \equiv \int \mathbf{R}' đ^3\mathbf{R}'$; later, we shall also make use of the third moment $\langle R^2\mathbf{R}\rangle \equiv \int (R')^2\mathbf{R}' đ^3\mathbf{R}'$.

The interaction of the nuclear monopole with the external electric field does not appear in the Hamiltonian as it corresponds to a mere constant offset. Interacting with the external field, the nuclear charge distribution gives rise to the dipole term $-e\langle \mathbf{R}\rangle \cdot \mathbf{E}_0$, while higher moments of $\rho(\mathbf{R})$ do not contribute when the external field $\mathbf{E}_0$ is uniform.

Step 2: Extract T-odd terms First we show that odd-parity moments, such as the EDM, violate both T and P. Any 2^ℓ-pole moment, for $\ell \in \mathbb{N}$, is a rank-ℓ tensor and as such must be aligned with a rank-ℓ tensor made from the spin operator $\mathbf{I}$. For example, the quadrupole moment is the rank-2 tensor $Q_q \propto T^2(\mathbf{I}, \mathbf{I})_q$. This explains why nuclear electric quadrupole moment is not exotic: it does not require P violation or T violation. Unlike in the case of a dipole moment, considered above, there are two $\mathbf{I}$'s, so T^2 is even under T-reversal. The $\ell = 3$ term, corresponding to an octupole moment, tends to be small, vanishing altogether in spin-½ nuclei, while the higher harmonics

are always negligible. [22]

We can proceed by looking for parity-odd terms in the Hamiltonian, knowing that such terms are also P, T-odd. One obvious parity-odd term is the dipole term

$$H_{\text{dip}} \equiv -Ze\langle \mathbf{R}\rangle \cdot \mathbf{E}_0, \tag{4}$$

explicitly present in $\hat{H}$. Further parity-odd terms can be extracted from $\hat{H}$ via the multipole expansion [23, eqn 3.38]

$$\frac{1}{|\mathbf{r}-\mathbf{R}|} = \sum_{\ell=0}^{\infty} \frac{r_<^\ell}{r_>^{\ell+1}} P_\ell(\hat{\mathbf{r}}\cdot\hat{\mathbf{R}}), \quad \begin{array}{l} r_< \equiv \min(r,R) \\ r_> \equiv \max(r,R) \end{array} \tag{5}$$

P_ℓ being the Legendre polynomials of degree ℓ, and $\hat{\mathbf{r}}$ and $\hat{\mathbf{R}}$ are the unit vectors $\mathbf{r}/r$ and $\mathbf{R}/R$. For completenes, we give here also the spherical multipole expansion [23, eqn 4.1],

$$\int \frac{\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} \mathrm{d}^3\mathbf{r}' = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \frac{4\pi}{2\ell+1} q_{\ell m} \frac{Y_{\ell m}(\theta,\varphi)}{r^{\ell+1}}, \tag{6}$$

where the multipole moment is $q_{\ell m} \equiv \int Y^*_{\ell m}(r')^\ell \rho(\mathbf{r}')\mathrm{d}^3\mathbf{r}'$.

For reasons stated in a previous paragraph, we need only consider the $\ell = 1$ part of the electron–nucleus term, yielding

$$H^{\ell=1}_{\text{e–n}} \equiv -Ze^2 \sum_i \mathbf{r}_i \cdot \left[\frac{1}{r_i^3} \int_0^{r_i} \mathbf{R}đ^3\mathbf{R} + \int_{r_i}^{\infty} \frac{\mathbf{R}}{R^3} đ^3\mathbf{R} \right]. \tag{7}$$

Let R_{N} be the radius at which $\rho(R_{\text{N}}) = 0$; i.e., the size of the nucleus. In the limit of a point-like nucleus, the second term above vanishes since there is no nuclear charge in the region $0 < R < r_i$, while for $r_i = 0$ both terms vanish anyway. The first term—all that is left of $H^{\ell=1}_{\text{e–n}}$—gives

$$\lim_{R_{\text{N}}\to 0} H^{\ell=1}_{\text{e–n}} = -Ze^2 \sum_i \frac{\mathbf{r}_i}{r_i^3} \cdot \langle \mathbf{R}\rangle. \tag{8}$$

We now conclude that the parity-odd moments of the nuclear charge distribution that give rise to P, T-violating interactions, consist of the following terms:

$$H_{\text{CPV}} \equiv H_{\text{dip}} + H^{\ell=1}_{\text{e–n}}. \tag{9}$$

In the next step, we shall see how these terms fail to cause a first-order energy shift in a point nucleus due to Schiff screening.

Step 3: Screening in point nucleus In this step, we show that any operator that can be written as a commutator with $\hat{H}$ will have a vanishing expectation value in any eigenstate of $\hat{H}$, and will thus cause no energy shift in the first order of perturbation theory. We then proceed to re-write H_{CPV} precisely in terms of such a commutator.

Let $|n\rangle$ be an eigenstate of the Hamiltonian $\hat{H}$ with energy E_n. Then the commutator of an arbitrary operator $\hat{O}$ with $\hat{H}$ has a zero expectation value in that state:

$$\begin{aligned} \langle n|[\hat{H},\hat{O}]|n\rangle &= \langle n|\hat{H}\hat{O}|n\rangle - \langle n|\hat{O}\hat{H}|n\rangle \\ &= E_n\langle n|\hat{O}|n\rangle - \langle n|\hat{O}|n\rangle E_n = 0. \end{aligned}$$

We now proceed to show that in a point-like nucleus, the external-field dipole term $-e\langle \mathbf{R}\rangle \cdot \mathbf{E}_0$, plus the T-odd part of the electron–nucleus interaction from eqn 7, is proportional to the commutator

$$\left[\textstyle\sum_i \mathbf{p}_i,\ \hat{H}\right] = \sum_i \left[\mathbf{p}_i,\ \left(-Ze^2 \int \frac{đ^3\mathbf{R}}{|\mathbf{r}_i - \mathbf{R}|} + e\mathbf{r}_i \cdot \mathbf{E}_0 \right) \right],$$

noting that the sum of momenta trivially commutes with the electron kinetic energy, the electron–electron interaction energy, and the energy of the nuclear dipole moment in the external field. The canonical commutation relation $[\mathbf{r}_i, \mathbf{p}_j] = i\hbar\delta_{ij}$ allows us to transform the above into

$$\left[\textstyle\sum_i \mathbf{p}_i,\ \hat{H}\right] = -Ze^2 \textstyle\sum_i (-i\hbar)\nabla_i \displaystyle\int \frac{đ^3\mathbf{R}}{|\mathbf{r}_i - \mathbf{R}|} + e\mathbf{E}_0 \sum_{i=1}^{Z} (-i\hbar).$$

Notice the sum in the second term, equal to $-i\hbar Z$. Factoring out the common factors, switching the order of the two terms, and pre-multiplying by $\langle \mathbf{R}\rangle$, we obtain

$$\begin{aligned} \langle \mathbf{R}\rangle \cdot \frac{\left[\sum_i \mathbf{p}_i,\ \hat{H}\right]}{i\hbar} &= -Ze\langle \mathbf{R}\rangle \cdot \left(\mathbf{E}_0 - e\textstyle\sum_i \nabla_i \displaystyle\int \frac{đ^3\mathbf{R}}{|\mathbf{r}_i - \mathbf{R}|} \right) \\ &= H_{\text{dip}} + H_{\text{comm}}, \end{aligned} \tag{10}$$

where we identify first term with H_{dip} from eqn 4, and name the second term H_{comm}. Expanding the latter using the multipole expansion, we note that the series can be terminated after the $\ell = 0$ term due to the fact that the higher harmonics are either T-even or negligible:

$$H_{\text{comm}} = \sum_\ell H^\ell_{\text{comm}} \approx H^{\ell=0}_{\text{comm}}. \tag{11}$$

The multipole formula (eqn 5) gives

$$H^{\ell=0}_{\text{comm}} = Ze^2 \sum_i \langle \mathbf{R}\rangle \cdot \nabla_i \left(\int_0^{r_i} \frac{đ^3\mathbf{R}}{r_i} + \int_{r_i}^{\infty} \frac{đ^3\mathbf{R}}{R} \right). \tag{12}$$

The ∇_i acts on the $\mathbf{r}_i$ coordinates and not on the $\mathbf{r}'$, killing the second *integrand* (not integral!). As the integration limits also contains r_i, we have to employ Leibniz's integral rule [24]

$$\frac{d}{dx}\left(\int_a^x f(x,t)\mathrm{d}t \right) = f(x,x) + \int_a^x \frac{\partial}{\partial x} f(x,t)\mathrm{d}t$$

In the case of eqn 12, the rule gives

$$H^{\ell=0}_{\text{comm}} = Ze^2 \sum_i \langle \mathbf{R}\rangle \cdot \left(\frac{1}{r_i} - \frac{1}{r_i} + \int_0^{r_i} \nabla_i \left(\frac{1}{r_i} \right) đ^3\mathbf{R} \right).$$

We see that the contributions from ∇_i acting on the limits of the two integrals are equal and opposite, and they cancel out. Noting that $\nabla_i(1/r_i) = -\mathbf{r}_i/r_i^3$, we finally obtain

$$H^{\ell=0}_{\text{comm}} = -Ze^2 \sum_i \mathbf{r}_i \cdot \frac{\langle \mathbf{R}\rangle}{r_i^3} \int_0^{r_i} đ^3\mathbf{r}'. \tag{13}$$

Recall that the nuclear charge density is normalized to 1. In the point-nucleus limit $R_N \to 0$, the above integral yields unity for any nonzero value of r_i, and the term is

$$\lim_{R_N \to 0} H^{\ell=0}_{\text{comm}} = -Ze^2 \sum_i \frac{\mathbf{r}_i}{r_i^3} \cdot \langle \mathbf{R} \rangle \tag{14}$$

This completes the proof of the Schiff theorem: the T-odd terms H_{CPV} in the Hamiltonian can be expressed as a commutator $[\hat{H}, \ldots]$ in the limit $R_N \to 0$. To reiterate:

$$\begin{aligned}
\lim_{R_N \to 0} H_{\text{CPV}} &= \lim_{R_N \to 0} \left(H_{\text{dip}} + H^{\ell=1}_{\text{e-n}} \right) && \text{(defined in eqn 9)} \\
&= \lim_{R_N \to 0} \left(H_{\text{dip}} + H^{\ell=1}_{\text{comm}} \right) && \text{(eqns 8 and 14)} \\
&\approx \lim_{R_N \to 0} \left(H_{\text{dip}} + H_{\text{comm}} \right) && \text{(as per eqn 11)} \\
&= \lim_{R_N \to 0} \left(\langle \mathbf{R} \rangle \cdot \frac{\left[\sum_i \mathbf{p}_i,\ \hat{H} \right]}{i\hbar} \right). && \text{(as per eqn 10)}
\end{aligned}$$

Thus H_{CPV} can cause no first order energy shift.

Step 4: Finite-size nucleus In heavy atoms, the limitation posed by Schiff's theorem is lifted by the finite size of the nucleus and the T-odd terms in the Hamiltonian can give rise to a measurable electrostatic interaction. Such terms can no longer be subsumed into a commutator with a zero expectation value. Consider the expression

$$\delta H \equiv H_{\text{CPV}} - \langle \mathbf{R} \rangle \cdot \frac{\left[\sum_i \mathbf{p}_i,\ \hat{H} \right]}{i\hbar}, \tag{15}$$

with the T-odd terms $H \equiv H_{\text{CPV}}$ as defined in eqn 9. The commutator always has a zero expectation value, and represents the part of the T-odd nuclear interaction that is fully screened by the electrons. The above difference thus stands for the residual interaction allowed by finite-size limitations of the Schiff theorem.

Note that this is not quite equivalent to adding a zero term—the commutator does have a zero expectation value, but not necessarily a zero matrix element. The expectation value of the commutator is zero even when there is an external E-field applied since that external field was in the Hamiltonian from the start. So, it is only the difference δH that can give a non-zero energy shift, under any circumstances, and we can ignore the commutator part.

As will be explained in the next step, the presence of an external electric field E_0, or a T-violating interaction δ_T, mixes the states into some superposition $|\psi\rangle \equiv |\text{s}\rangle + \eta|\text{p}\rangle$ of the electronic s and p states. We now show that when both E_0 and H_{CPV} are nonzero, the system acquires an energy shift that depends on the sign of E_0, and on the magnitude of the T-violating term in the Hamiltonian. Evaluating the expectation value of δH in that superposition state, we find that

$$\langle \delta H \rangle \equiv \langle \psi | \delta H | \psi \rangle = 2\eta \langle \text{s} | \delta H | \text{p} \rangle; \tag{16}$$

the diagonal elements vanish by parity. It is not sufficient merely to have the interaction δH; the atom also needs to be polarized so that $\eta \neq 0$. Indeed, there are two equivalent ways to arrive at an energy shift proportional to the field:

- In the absence of an external field, δH causes no first order shift. It does, however, give rise to a minute $\eta \neq 0$. Were a large external electric field applied as well, there would be an energy shift proportional to η and the field. This is explained below in detail.
- In the absence of δH, it is the external field $\mathcal{E}$ that causes a mixing $\eta \propto E$ between $|\text{s}\rangle$ and $|\text{p}\rangle$ states. If the nucleus does, after all, have a nonzero δH, it will cause a linear energy shift as per eqn 16.

In the following two steps, we see how either the external field, or an off-diagonal interaction, can cause a mixing η between opposite-parity states (step 5). We shall see that the resulting energy shift in an external field is indeed proportional to η. Finally, we evaluate the matrix elements of δH between the opposite-parity states (step 6).

Step 5: Polarization in atoms Consider a two-level atom with the states $|\text{s}\rangle$ and $|\text{p}\rangle$ of opposite parity (Fig. 1a). The two states are separated in energy by Δ_{sp}, and coupled by the dipole interaction $D_{\text{sp}}\mathcal{E}$, and an additional small coupling δ_T from new, T-violating physics. The Hamiltonian matrix is

$$H = \begin{pmatrix} 0 & x \\ x & \Delta_{\text{sp}} \end{pmatrix}, \quad x \equiv D_{\text{sp}}\mathcal{E} + \delta_T. \tag{17}$$

Diagonalizing the Hamiltonian and expanding around small off-diagonal coupling x, we get the eigenvalues

$$E_\pm = \frac{1}{2}\left(\Delta_{\text{sp}} \pm \sqrt{4x^2 + \Delta_{\text{sp}}^2} \right) \approx \begin{cases} \Delta_{\text{sp}} + x^2/\Delta_{\text{sp}}, & \text{for } E_+ \\ \quad\ \ - x^2/\Delta_{\text{sp}}, & \text{for } E_-. \end{cases}$$

In the opposite limit ($x \gg \Delta_{\text{sp}}$), the energies are

$$\begin{aligned} E_\pm &\approx \pm\sqrt{x^2} \\ &\approx \pm\big(|D_{\text{sp}}\mathcal{E}| + \text{sgn}(D_{\text{sp}}\mathcal{E})\delta_T \big); \end{aligned} \tag{18}$$

the second approximate equality holds for $|\delta_T| \ll |D_{\text{sp}}\mathcal{E}|$.

We see that placing the atom in an external field, in the absence of a T-violating term δ_T, does not cause a first order energy shift. What does happen instead is that the states mix. The exact eigenstates are

$$|\psi_\pm\rangle = N^{-1/2} \left[|\text{s}\rangle - \frac{2x}{\Delta_{\text{sp}} \pm \sqrt{4x^2 + \Delta_{\text{sp}}^2}} |\text{p}\rangle \right], \tag{19}$$

with N a normalizing constant. Expanding the eigenvectors $|\psi_\pm\rangle$ in the limit $x \ll \Delta_{\text{sp}}$ to first order, we obtain

$$|\text{s}\rangle - \frac{D_{\text{sp}}\mathcal{E} + \delta_T}{\Delta_{\text{sp}}} |\text{p}\rangle \quad \text{and} \quad |\text{p}\rangle + \frac{D_{\text{sp}}\mathcal{E} + \delta_T}{\Delta_{\text{sp}}} |\text{s}\rangle,$$

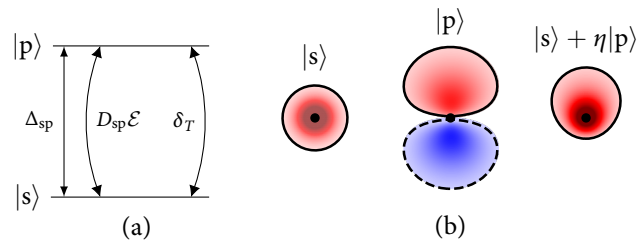

Figure 1: ***a.*** Level diagram for the two-state atom polarization model. ***b.*** Distortion of atomic shape due to polarization. The black dot indicates the nucleus / origin of coordinates. The wavefunctions shown are, to within normalization, the hydrogenic $\psi_s \propto e^{-r}$ and $\psi_p \propto re^{-r/2}\cos\theta$, shown in arbitrary radial units and phase but otherwise to scale. Contours placed at $\psi = 0.015$; negative wavefunction indicated by dashed contour. The states on the right are mixed by $\eta = 0.5$ for emphasis; see text for physical estimates of η.

whereas in the opposite limit ($x \gg \Delta_{sp}$), of course, the eigenstates are $2^{-1/2}(|s\rangle \pm |p\rangle)$.

Physically, the electron cloud gets repelled by the external field, and the nucleus attracted towards it; the change of shape matches what we expect from the combined wavefunction (Fig. 1b). We say that the atom has become polarized. To describe the extent of the polarization, define

$$\mathcal{P} \equiv \frac{\text{induced dipole}}{\langle s|\hat{d}|p\rangle} = \frac{\langle \psi_\pm|\hat{d}|\psi_\pm\rangle}{D_{sp}} \xrightarrow{\text{small field}} 2\frac{D_{sp}\mathcal{E} + \delta_T}{\Delta_{sp}},$$

noting that the dipole operator $\hat{d}$ only has non-zero matrix elements (D_{sp}) between states of opposite parity. In general, for the eigenstates $N^{-1/2}(|s\rangle + \eta|p\rangle)$ of the two-state Hamiltonian, with N a normalizing constant, the polarization will be $2\eta/N$. See above for the exact expression of the mixing coefficient η (eqn 19), as well as the approximate forms valid in the high- and low-field limits.

This means that either the external field, or the T-violating interactions, can both polarize the system. The small-field (alternatively, small x) energy shift can be written in $\mathcal{P}$ as

$$\Delta E = \frac{\mathcal{P}}{2}\left(D_{sp}\mathcal{E} + \delta_T\right).$$

In the opposite limit, where $\mathcal{P} \approx 1$, the energy shift is also proportional to δ_T and $\operatorname{sgn}\mathcal{E}$ as per eqn 18.

Of course, as stated in the previous step, the argument can be inverted: in the absence of an external field, δ_T polarizes the atom, and an external field is necessary for a shift linear in both δ_T and $\mathcal{E}$. Nuclear T-odd interactions, and the external field, are not the only thing that can polarize an atom—an electron EDM can have the same result. Indeed, so can anything that can couple the two states of opposite parity with a real matrix element. By contrast, ordinary electroweak interactions mix $|s\rangle$ and $|p\rangle$, but with pure imaginary matrix element.

Step 6: Matrix elements We now evaluate the matrix elements $\delta_T = \langle s|\delta H|p\rangle$. Inserting eqns 9, 7, 10, 11 and 13 into eqn 15, we can write out the interaction explicitly:

$$\frac{\delta H}{-Ze^2} = \sum_i \mathbf{r}_i \cdot \left[\frac{1}{r_i^3}\int_0^{r_i} \mathbf{R}đ^3\mathbf{R} + \int_{r_i}^{\infty}\frac{\mathbf{R}}{R^3}đ^3\mathbf{R} - \frac{\langle\mathbf{R}\rangle}{r_i^3}\int_0^{r_i} đ^3\mathbf{R}\right].$$

Instead of summing over Z discrete electrons, the electron density is described by the product of appropriate wavefunctions. Note that we now consider the effect of only a single electron, as indicated by the change in notation $r_i \to r$. The sum $\sum_i \mathbf{r}_i$ becomes the integral $\int (r\hat{\mathbf{r}})\psi_s^*\psi_p d^3\mathbf{r}$:

$$\langle s|\delta H|p\rangle = -Ze^2\int r(\mathbf{S}'\cdot\hat{\mathbf{r}})(\psi_s^*\psi_p)_{r\to 0}d^3\mathbf{r}, \tag{20}$$

where we define the vector

$$\mathbf{S}'(\mathbf{r}) \equiv \frac{1}{r^3}\int_0^r \mathbf{R}đ^3\mathbf{R} + \int_r^\infty \frac{\mathbf{R}}{R^3}đ^3\mathbf{R} - \frac{\langle\mathbf{R}\rangle}{r^3}\int_0^r đ^3\mathbf{R}.$$

Observe that the vector $\mathbf{S}'(\mathbf{r})$ vanishes for $r > R_N$. On the scale of electronic wavefunctions, the above is nonzero only near the origin since $R_N \ll a_0$, hence the $r \to 0$ subscript on the product of the wavefunctions in 20.

Now group together the terms with the same limits:

$$\mathbf{S}'(\mathbf{r}) = \int_0^r \frac{\mathbf{R} - \langle\mathbf{R}\rangle}{r^3}đ^3\mathbf{R} + \int_r^\infty \frac{\mathbf{R}}{R^3}đ^3\mathbf{R}. \tag{21}$$

The difference in the first integral vanishes in the limit $r \to \infty$ where $\int_0^r đ^3\mathbf{R}$ can be replaced by $\int_0^\infty đ^3\mathbf{R}$ whenever $r > R_N$:

$$\lim_{r\to\infty}\left[\int_0^r \mathbf{R}đ^3\mathbf{R} - \langle\mathbf{R}\rangle\int_0^r đ^3\mathbf{R}\right] = \langle\mathbf{R}\rangle - \langle\mathbf{R}\rangle\int_0^\infty đ^3\mathbf{R} = 0,$$

so we can change the limits of the first integral in 21 as follows:

$$\int_0^r \frac{\mathbf{R} - \langle\mathbf{R}\rangle}{r^3}đ^3\mathbf{R} = \int_0^\infty \frac{\mathbf{R} - \langle\mathbf{R}\rangle}{r^3}đ^3\mathbf{R} - \int_r^\infty \frac{\mathbf{R} - \langle\mathbf{R}\rangle}{r^3}đ^3\mathbf{R}.$$

The first term on the right-hand side vanishes, so that eqn 21 leaves us with three terms with the same limits:

$$\mathbf{S}'(\mathbf{r}) = \int_r^\infty \left(\frac{\mathbf{R}}{R^3} - \frac{\mathbf{R}}{r^3} + \frac{\langle\mathbf{R}\rangle}{r^3}\right)đ^3\mathbf{R}. \tag{22}$$

The product of wavefunctions can be separated into a product of angular and radial functions: $\psi_s^*\psi_p = U(\mathbf{\Omega})\sum_{k=1}^\infty b_k r^k$, where $\mathbf{\Omega}$ stands for the angular coordinates θ, φ. In the non-relativistic case, only b_1 is nonzero [22]. This allows us to write the product of wavefunctions near the origin as

$$\psi_s^*\psi_p = r\frac{\psi_s^*\psi_p}{r} = r\frac{\partial}{\partial r}(\psi_s^*\psi_p) = r(\hat{\mathbf{r}}\cdot\nabla)(\psi_s^*\psi_p). \tag{23}$$

With $\mathbf{S}'$ from eqn 22 and wavefunctions from eqn 23, we can separate eqn 20 as follows:

$$\langle \mathrm{s}|\delta H|\mathrm{p}\rangle = 3e \int \mathrm{d}^2\mathbf{\Omega}(\hat{\mathbf{r}}\cdot\nabla)(\psi_\mathrm{s}^*\psi_\mathrm{p})_{r\to 0}\hat{\mathbf{r}}\cdot\mathbf{S}, \tag{24}$$

placing some constants next to the angular factors, and others in the radial part $\mathbf{S}$, defined as

$$\mathbf{S} \equiv -\frac{Ze}{3}\int_0^\infty r^4 \mathrm{d}r \int_r^\infty \left(\frac{\mathbf{R}}{R^3} - \frac{\mathbf{R}}{r^3} + \frac{\langle\mathbf{R}\rangle}{r^3}\right) đ^3\mathbf{R},$$

where the r^4 consists of r^2 from the volume element d^3r, a factor of r from the wavefunctions, and another that is already present in eqn 20. We have to change the order of integration:

$$\mathbf{S} = -\frac{Ze}{3}\int_0^\infty đ^3\mathbf{R} \int_0^R r^4 \left(\frac{\mathbf{R}}{R^3} - \frac{\mathbf{R}}{r^3} + \frac{\langle\mathbf{R}\rangle}{r^3}\right) \mathrm{d}r,$$

which integrates to

$$\mathbf{S} = \frac{Ze}{10}\left[\langle R^2\mathbf{R}\rangle - \frac{5}{3}\langle R^2\rangle\langle\mathbf{R}\rangle\right], \tag{25}$$

This is the usual expression for the Schiff moment [25, eqn 29] save for the additional factor of Z reflecting the different normalization (to unity instead of Z) we have chosen for the nuclear charge distribution $\rho(\mathbf{r})$.

To evaluate the angular part in eqn 24, we first extract two factors of $\cos\theta$ from the two dot products in the integrand:

$$\hat{\mathbf{r}}\cdot\mathbf{S} = S\cos\theta$$
$$\hat{\mathbf{r}}\cdot\nabla = \frac{\partial}{\partial r} = \cos\theta\frac{\partial}{\partial(r\cos\theta)} = \cos\theta\frac{\partial}{\partial z} = \cos\theta(\hat{\mathbf{z}}\cdot\nabla),$$

where $\mathbf{S} = S\hat{\mathbf{z}}$. Thus the equation becomes

$$\begin{aligned}\langle \mathrm{s}|\delta H|\mathrm{p}\rangle &= \left(3e\int\cos^2\theta \mathrm{d}^2\mathbf{\Omega}\right)\mathbf{S}\cdot\nabla(\psi_\mathrm{s}^*\psi_\mathrm{p})_{r\to 0}\\ &= 4\pi e\,\mathbf{S}\cdot\nabla(\psi_\mathrm{s}^*\psi_\mathrm{p})_{r\to 0},\end{aligned} \tag{26}$$

reproducing eqn 14 in ref. [22]. Trivially, the effective Schiff potential $\varphi(\mathbf{r}) \equiv -4\pi\,\mathbf{S}\cdot\nabla\delta^3(\mathbf{r})$ will have the same matrix elements $\langle \mathrm{s}| - e\varphi(\mathbf{r})|\mathrm{p}\rangle$ as in eqn 26.

Knowing the gradient $\nabla(\psi_\mathrm{s}^*\psi_\mathrm{p})$ of the electron probability density near the nucleus, we can use eqn 26 to relate the measured energy shifts with the expected magnitude of the Schiff moment. As a crude approximation, we can take

$$\nabla(\psi_\mathrm{s}^*\psi_\mathrm{p}) = \frac{Z^2}{a_0^4} \quad\to\quad S = \frac{\nu}{3.65\cdot 10^6}\frac{e\,\mathrm{fm}^3}{\mathrm{Hz}}, \tag{27}$$

using $Z = 81$ for thallium. More precisely, $\nabla(\psi_\mathrm{s}^*\psi_\mathrm{p}) \approx 1016/a_0^4$ as per ref. [26], so that, as in eqn 26 of ref. [27], we have

$$S = \frac{\nu}{5.67\cdot 10^5}\frac{e\,\mathrm{fm}^3}{\mathrm{Hz}}. \tag{28}$$

Note, however, that there is a significant uncertainty about this number. For instance, a more recent calculation [28, eqn 22] gives the number in the denumerator above as $1.98\cdot 10^6$.

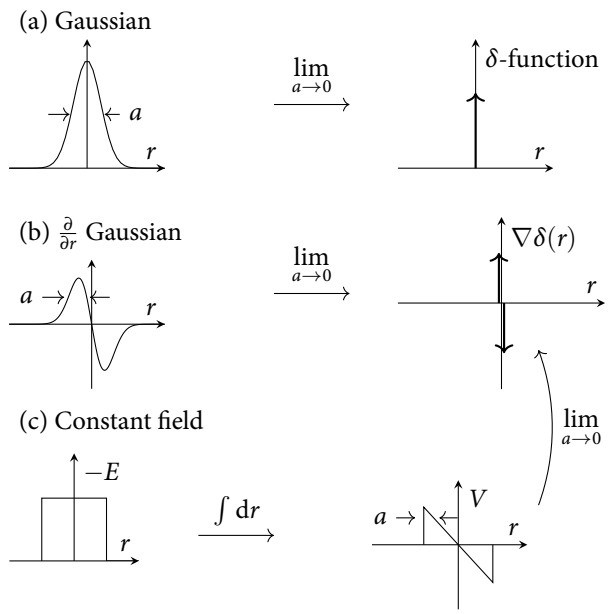

Figure 2: ***a.*** Delta function as the limit of a Gaussian. ***b.*** Gradient of a Gaussian as a representation of $\nabla\delta(r)$. ***c.*** Constant field inside the nucleus gives rise to a truncated linear potential, which has a similar shape to the derivative of the Gaussian and in the appropriate limit becomes $\nabla\delta(r)$.

Step 7: Interpretation To get an intuitive grasp on what the expression $\nabla\delta^3(\mathbf{r})$ means, consider the definition of the delta function as the limit of a Gaussian distribution,

$$\delta(r) = \lim_{a\to 0}(a\sqrt{\pi})^{-1}e^{-r^2/a^2},$$

as in Fig. 2a. Taking the derivative of the Gaussian, we get a shape that is negative for $r < 0$, positive for $r > 0$, zero at $r = 0$ and also vanishing for $r \to \pm\infty$. That shape (Fig. 2b), stretched to infinity, is then representative of $\nabla\delta(\mathbf{r})$.

We may think of the Schiff moment as due to a constant electric field along the direction of the nuclear spin $\hat{I}$ inside a nucleus of radius R_0. This potential is linear in the coordinate r and truncated outside of the nucleus (i.e., at $|r| > R_0$). In the limit of a small nucleus, this will have a similar shape (Fig. 2c) as the expression $\nabla\delta(\mathbf{R}_0)$ discussed in the preceding paragraph.

It is possible to estimate the magnitude of the Schiff moment inside a nucleus with a dipole moment d_N and a uniform spherical core of radius R_0. The EDM includes additive contributions from both the nucleon EDMs, arising from short-range interactions between quarks, and also from longer-range P, T-violating interactions between valence nucleons and the core. In fact, in some theory models, the latter contribution to the SMT can be 10 to 100 times larger than that due to the nucleonic EDM. [25] Consider models for each case:

(1) Inside the uniform charge distribution—which does not contribute to the SMT—there are in addition two point charges, separated by a distance $2R_0$ along the z axis. The T-odd part of

the nuclear charge can thus be written as

$$\rho(\mathbf{R}) = \frac{d_{\mathrm{N}}}{2eZR_0}\left[\delta^3(\mathbf{R} - R_0\hat{\mathbf{z}}) - \delta^3(\mathbf{R} + R_0\hat{\mathbf{z}})\right],$$

the constant factors being chosen so that $e\langle\mathbf{R}\rangle = d_{\mathrm{N}}\hat{\mathbf{z}}$. Evaluating the third moments in eqn 25, we readily see that the magnitude of the SMt is $|\mathbf{S}| = d_{\mathrm{N}}R_0^2/10$.

(2) Instead of the two separated point charges, suppose the nucleus consists of closed shells, plus a valence proton (or neutron) that carries a dipole moment d_{p}. Further suppose that the valence nucleon is distributed uniformly throughout the nucleus, so that the nucleus has uniformly distributed electric polarization. It is a standard result that the charge distribution of a uniformly polarized sphere is exactly a surface charge, with charge density

$$\rho(\mathbf{R}) = \frac{d_{\mathrm{p}}}{4\pi eZR_0^3/3}\delta(R_0 - R)\cos\theta,$$

which integrates to $e\langle\mathbf{R}\rangle = d_{\mathrm{p}}\hat{\mathbf{z}}$, and the Schiff moment is $|\mathbf{S}| = d_{\mathrm{p}}R_0^2/10$. We have seen that the effect of this electric field at the site of the nucleus would be to polarize the atom, and give rise to a potentially measurable energy shift.

In each case, we saw that $|\mathbf{S}| \propto R_0^2$. Furthermore, the matrix element (eqn 27) proportional to the first order energy shift has a Z^2 dependence. Since the nuclear radius scales as $R_0 \approx R_{\mathrm{N}}A^{1/3}$, where $R_{\mathrm{N}} \approx 1.2$ fm is the characteristic nuclear size [29, eqn 3.8], the overall scaling is $Z^2A^{2/3}$. We conclude that heavier atoms—such as thallium—are preferable for SMt experiments.

C. New physics in molecules

For nuclei inside a molecule, the Schiff moment and other P, T-violating effects give rise to an effective interaction of form

$$\mathcal{H} = -\nu\hat{\mathbf{I}}\cdot\hat{\mathbf{n}}, \tag{29}$$

where $\hbar\hat{\mathbf{I}}/2$ is the nuclear spin, and $\hat{\mathbf{n}}$ is a unit vector along the internuclear axis in the molecule (in TlF, we define this to point towards the F nucleus). Compared to the atomic case, in a diatomic molecule there is another inherent direction: along the internal electric field, pointing along the internuclear axis. The interaction strength ν can then be interpreted in terms of a molecular EDM, proton EDM, Schiff moment, or other quantities.

Without an external electric field, the interaction of eqn 29 fails to produce any first order effects in any given rotational state. According to [30, 31], this is because the Tl nuclear spin is perpendicular to the internuclear axis in an unpolarized molecule. Stated more correctly, the rotation of the molecule averages $\hat{\mathbf{n}}$ to zero, and the Schiff shift—proportional to the internal molecular field—vanishes. [32, 27] However, if we apply an external field, the molecule becomes polarized, which destroys the inversion symmetry and a nonzero expectation results.

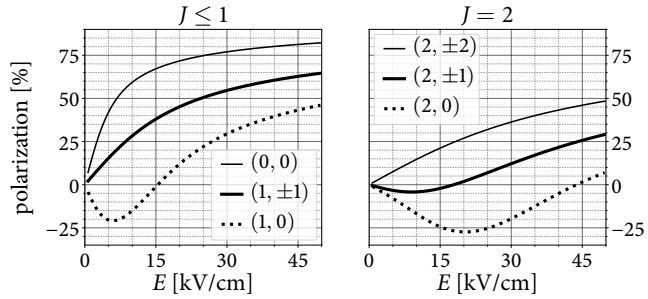

Figure 3: Degree of polarization of TlF molecules in an external field. The legends give the labels (J, m_J). Note that J is not a good quantum number at high fields and is to be understood in an asymptotic sense.

Polarization Compared to atoms, molecules can be polarized much more readily in laboratory fields owing to their smaller level separation (tens of GHz, or 10^{-4} eV, instead of the typical atomic eV-scale spacing), making them ideal for an EDM measurement. Thus Sandars [33] suggested a molecular-beam resonance experiment could be used to probe the existence of an electric dipole moment on the proton, so long as the molecule has a heavy atom with an unpaired proton in the nucleus.

The polarization of molecules is so effective that the first order model, entirely suitable for atoms, no longer applies. When the coupling $D_{\mathrm{sp}}\mathcal{E} + \delta_T$ is no longer small compared to the level separation, the states get mixed with a mixing coefficient of order unity. It is customary to define the degree of polarization $\mathcal{P}$ of a state ψ as

$$\mathcal{P} \equiv \frac{\langle\hat{d}_z\rangle}{D_{\mathrm{sp}}} = \langle\cos\theta\rangle, \tag{30}$$

where the expectation values $\langle\cdots\rangle$ are of course evaluated as $\langle\psi|\cdots|\psi\rangle$. See Fig. 3 for a plot of $\mathcal{P}$ in some of the lowest-lying rotational states of TlF; observe how states that are lower in energy are easier to polarize, while some of the higher-lying states can even be polarized against the direction of the applied field at some field magnitudes. Despite the fact that the $J = 0$ states can be polarized better than the $J = 1$, $m_J = \pm 1$ ones, we prefer the latter due to their internal field, nearly exactly aligned with the applied field. The former states woud necessitate applying an external magnetic field, suffering from the motional magnetic field systematic (discussed in section II).

As we have seen previously, a T-violating interaction such as the Schiff moment inside a fully polarized molecule gives rise to a first order energy shift $h\nu$ as in eqn 28. We have also seen two simple models of how a dipole moment d inside the nucleus results in a SMt of magnitude $S = ZdR_0^2/10$. We may thus estimate the size of an effective internal molecular electric field $\mathcal{E}_{\mathrm{int}}^{\mathrm{eff}}$ via the definition

$$\mathcal{E}_{\mathrm{int}}^{\mathrm{eff}} \equiv \frac{h\nu}{d} = 5.67\cdot 10^5\,\frac{h\ \mathrm{Hz}}{e\ \mathrm{fm}^3}\,\frac{R_0^2}{10} = 117\ \mathrm{kV/cm}. \tag{31}$$

We emphasize that this is not the average field in the volume of

the nucleus; that field must be zero to keep the nucleus from accelerating out of the molecule. Nor is $\mathcal{E}_{\rm int}^{\rm eff}$ to be taken as a proxy for the simple $\sim$ GV/cm field due to the nuclear charge at a distance of order a_0 from the nucleus. Furthermore, the numerical values in eqn 31 obtain hugely different estimates in different papers, nearly three orders of magnitude (from 10 kV/cm to 10 MV/cm). Trying to estimate a generalized form of the SMt, Flambaum and Ginges [22] found the "analytical calculation hopeless [for] a nucleus with an external proton in state $s_{1/2}$, as is the case for ^{203,205}Tl."

Separated oscillatory fields To obtain an upper bound on v, a pair of states is chosen such that one of the states moves up in energy due to the interaction in eqn 29, and the other one down by the same amount, increasing the level separation by $2v$. Upon reversal of externally applied fields, the directions of the level shifts change and a level separation of $-2v$ obtains compared to the levels in absence of the interaction. (See Fig. 1 of ref. [30] or [31] for a diagram.) Comparing the two perturbations of the level spacing, a $4v$ difference (together with its error bars) can be quoted as the main result of molecular-beam symmetry-violation experiments.

The spacing between the levels paragraph can be measured by a technique based on Rabi oscillations. Consider two states, $|p\rangle$ and $|q\rangle$, separated by an energy $\hbar\omega_0$, and placed in an oscillating field $\hbar b e^{-i\omega\tau}$. Defining the detuning $\delta \equiv \omega_0 - \omega$, the population transfer probability is given by [34, eqn V.10]

$$P_{p,q} = \left(\frac{2b}{\Omega}\right)^2 \sin^2\left(\frac{\Omega\tau}{2}\right), \text{ with } \Omega = \sqrt{\delta^2 + (2b)^2}.$$

We see the transfer is maximised for a resonant field, $\omega = \omega_0$, and the full width at half maximum (FWHM) is given by $\Delta\omega = 4b$. For a resonant $\pi/2$ pulse of duration $T = \pi/2b$, the FWHM will be $\Delta\omega = 2\pi/T$, and the longer the precession period T, the more precisely we can determine the resonant frequency ω_0.

In principle, an infinitely precise measurement could be obtained by simply extending the length l of the precession region, thus prolonging the precession period $\tau = l/v$ for molecules of speed v. In practice, however, this meets with the following problems. (1) The beam intensity N decays as $1/l^2$ in the limit where the interaction region is long compared to the distance from the source to the start of the interaction region. Indeed, the signal-to-noise ratio, being proportional to $\sqrt{N}/\tau$, is roughly independent of interaction length. (2) We may want to measure the precession in a region where the oscillating field cannot be applied. (3) It is difficult to maintain uniform externally applied fields over an extended length.

It turns out that the oscillating field does not need to extend over the entire length of the precession region. In Ramsey's resonance scheme, separated oscillating fields (SOF) are only applied in two small regions at the beginning and at the end of the interaction region, each applying a $\pi/2$ pulse. The resulting FWHM is about 40% smaller that that obtained in a simple

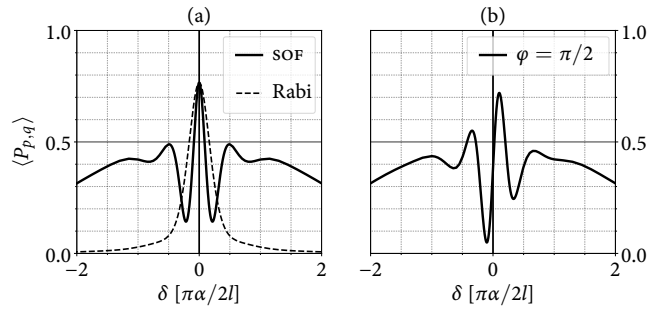

Figure 4: Transition probability as a function of detuning δ, averaged over the effusive-source speed distribution $v^3 e^{-v^2/\alpha^2}$, for a precession length $L = 10l$. ***a.*** Comparing SOF ($\tilde{b}_{\rm SOF} = 0.600$, $\varphi = 0$) and Rabi oscillations ($\tilde{b}_{\rm Rabi} = 1.200$), both with the same L. Notice the $\approx 40\%$ narrower FWHM for SOF. ***b.*** Detection scheme sensitive to small δ: SOF with $\varphi = \pi/2$ with the same $L = 10l$.

Rabi oscillation scheme, and the resonances are less broadened by the non-uniformity of the fixed field. [35] The transition probability $\widetilde{P}_{p,q}$, here expressed in terms of the simple Rabi oscillation probability $P_{p,q}$, is [34, cf. eqn V.37]

$$\frac{\widetilde{P}_{p,q}}{P_{p,q}} = 4\left[\cos\frac{T\overline{\delta}-\varphi}{2}\cos\frac{\Omega\tau}{2} - \frac{\delta}{\Omega}\sin\frac{T\overline{\delta}-\varphi}{2}\sin\frac{\Omega\tau}{2}\right]^2,$$

where $\overline{\delta} \equiv \overline{\omega}_0 - \omega$ is the detuning of the oscillating field frequency ω from the hyperfine level separation $\hbar\overline{\omega}_0$, averaged over the length of the interaction region, and φ is the phase delay between the two oscillating coils. Now T is the duration of the molecular flight through the field-free region, while τ is the duration of each of the oscillating-field regions. Of course, as a molecule flies through the different regions, the oscillating field is not turned off or on instantaneously; nevertheless, the effect of the smooth turn-on if simply a modified strength of the oscillating field, for which we compensate by directly optimizing the field strength.

Schropp [36] draws a parallel between the method of SOF, where a frequency-domain interference pattern is caused by two coils that are separated in times, and Young's double-slit experiment, where the distance between two slits causes a spatial interference pattern. In either case, the linewidth is inversely proportional to the field-free separation, whether a spatial one (double slits) or a temporal one (SOF).

The above transition probabilities hold for the case of molecules subjected to the specified interactions for a fixed amount of time. However, in a realistic molecular beam there is a distribution of velocities v and a corresponding distribution of times $\tau = l/v$ the perturbations act. We first consider the normalized probability density function corresponding to effusive beam sources, as used in the classic paper by Ramsey et al. [31]:

$$f(v) = \frac{2}{\alpha^4} v^3 e^{-v^2/\alpha^2}, \quad \alpha = \sqrt{\frac{2kT}{m}}, \tag{32}$$

which is simply a Boltzmann distribution with an extra factor

of v to account for the fact that faster molecules preferentially escape from the cell. The parameter α is the most probable molecular velocity, as well as a measure of the spread of velocities.

The Rabi oscillation probability $P_{p,q}$ can thus be averaged over the chosen distribution of molecular speeds $f(v)$ as

$$\langle P_{p,q}\rangle \equiv \int_0^\infty P_{p,q} f(v) dv. \tag{33}$$

The optimal choice of the oscillation strength b is directly related to the length of the resonance region l and the molecular velocity (or velocity spread) α, making $\pi\alpha/2l$ a convenient choice of frequency units. Thus we put $b = \tilde{b}(\pi\alpha/2l)$, defining the dimensionless oscillation strength $\tilde{b}$, and analogously for $\tilde{\delta}$ and $\tilde{\Omega}$. The integral in eqn 32 can be expressed in terms of a Meijer G-function [37, sect 5.3] as

$$\langle P_{p,q}\rangle = \frac{2\tilde{b}^2}{\tilde{\Omega}^2} - \frac{\pi^{5/2}\tilde{b}^2}{8} G^{2,0}_{0,3}\left(\frac{\pi^2\tilde{\Omega}^2}{16}\middle| -1, 1, -\frac{1}{2}\right)$$

The G-function can be evaluated numerically; in particular, we find that $\langle P_{p,q}\rangle(\tilde{\delta} = 0)$ is maximised for $\tilde{b} = 1.200$.

The distribution-averaged form of the SOF probability $\widetilde{P}_{p,q}$ can also be found in terms of special functions, but is considerably more complicated. In addition to the length l of the two oscillating coils, there is a further dependence on the free-precession length L, as well as the phase delay φ between the two coils. If the coils oscillate in phase ($\varphi = 0$), the peak of the transition probability is centered around the resonant frequency with no linear dependence on the detuning δ. Thus SOF experiments commonly take $\varphi = \pm\pi/2$, where

$$\langle\widetilde{P}_{p,q}\rangle \approx 0.383 \pm (0.599 + 0.495L/l)\tilde{\delta} + O(\tilde{\delta}^2);$$

the two choices of φ are depicted in Fig. 4. The above numbers assume $\tilde{b} = 0.600$, which is the optimal value of $\tilde{b}$ for separated oscillatory fields in the sense of maximum on-resonance transition probability (for $\varphi = 0$), or maximum slope of $\langle\widetilde{P}_{p,q}\rangle$ around the resonance (for $\varphi = \pi/2$). We see that the slope is directly proportional to the length of the free-precession region, as expected.

It will be noticed that there are high-slope regions in Fig. 4 for both $\varphi = 0$ and $\varphi = \pm\pi/2$, however for $\varphi = \pm\pi/2$, the maximum slope region appears at zero detuning. Due to a pulse-to-pulse variation in the beam properties, the exact width of the resonance (and thus the position of the highest-slope part of the $\varphi = 0$ curve) would change over time, whereas a zero detuning always occurs in the same place despite the beam parameters, and only the slope itself changes.

For a cryogenic buffer-gas beam source, the appropriate averaging is to be taken over an offset-Gaussian distribution of molecular velocities:

$$\langle\widetilde{P}_{p,q}\rangle_{\mathrm{BG}} \equiv \int_{-\infty}^{\infty} P_{p,q} f_{\mathrm{BG}}(v) dv, \quad f_{\mathrm{BG}}(v) = \frac{e^{-(v-\bar{v})^2/2\alpha^2}}{\sqrt{2\pi}\alpha},$$

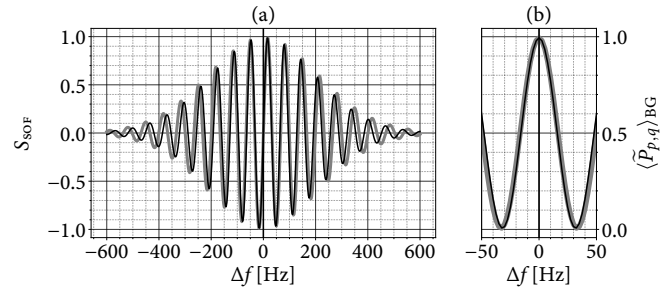

Figure 5: Buffer-gas SOF signal as a function of detuning Δf in linear units, averaged over a Gaussian speed distribution $\exp[-(v-\bar{v})^2/2\alpha^2]$ with $\bar{v} = 200$ m/s and $\alpha = 10$ m/s, for a 3 m separation between the oscillatory coils, and field strength $b = 9.975\,\pi\alpha/2l$. Observe the effect of the finite coil length: $l = 1$ cm (thick gray) and $l = 10$ cm (thin black). ***a.*** Difference in the averaged probabilities between $\varphi = \pi/2$ and $\varphi = -\pi/2$. ***b.*** Averaged transition probability for $\varphi = 0$ (detail).

where $\bar{v}$ is the average (center of mass) velocity of the molecular beam, with a spread of velocities α. The extended integration limits compared with eqn 33 are not significant, since the $f_{\mathrm{BG}}(v)$ for v far away from $\bar{v}$. See Fig. 5 for typical parameter values. Since the measurement is repeated for each sign of $\varphi = \pm\pi/2$, we can define the signal S_{SOF} as

$$S_{\mathrm{SOF}} = \langle\widetilde{P}_{p,q}\rangle_{\mathrm{BG}}(\varphi = \pi/2) - \langle\widetilde{P}_{p,q}\rangle_{\mathrm{BG}}(\varphi = -\pi/2),$$

as displayed in Fig. 5a. For the parameter values in that figure, the slope of S_{SOF} around zero detuning is approx. 0.094/Hz.

Beam sources The earliest molecular beams of thallium fluoride were of a thermal design, where a mass of the solid salt is heated in a copper oven to raise the vapour pressure. Depending on the mean free path (m.f.p.) of the molecules in the oven, we can distinguish two main types of thermal sources. In *effusive* sources, the m.f.p. is comparable with the linear dimension d of the hole through which the molecules emerge from the beam source. The particles escape by chance, and the beam properties reflect the state of the molecules in the oven. By contrast, in *jet sources*, the m.f.p. is much smaller than d. The beam properties no longer reflect those of the oven molecules; instead, the local thermal (electronic, rotational, and vibrational) is partly converted into the center-of-mass motion of the particles. These beams are thus faster but colder.

@Dave: I'm not sure it makes sense to have a table of beam sources, since I only present a couple beam sources and they each have a different set of interesting parameters. Unless there's lots of other beam sources that I should present?

For example, Schropp et al. [38, 36] report having a source at temperature 380 °C that yields a fraction 3×10^{-4} of molecules in the desired state $J = 1$, $m_J = 0$, out of a detected beam of 7.5×10^9 molecules/second. The beam velocity can be estimated as $v_{\mathrm{CM}} = \sqrt{2kT/m} \approx 230$ m/s, with a spread of order $v_{\mathrm{CM}}\sqrt{\ln 2} \approx 190$ m/s.

At the oven temperature of 380 °C, we expect approximately 25% of molecules to exist as a monomer, the rest being mostly dimers $(TlF)_2$. In addition to decreasing the useful signal, dimers were twice as effective at producing background counts with the detection technology used. However, by Le Chatelier's principle the equilibrium of the endothermic reaction $(TlF)_2 \rightleftharpoons 2TlF$ can be shifted towards the right by an increase in temperature or a decrease in pressure. Thus the oven consists of two stages: a lower, high-pressure section with the boiling salt, and an uppper, lower-pressure region, separated by a constriction. As a result, the monomer fraction may be as high as 50% in the two-chamber design.

As an example of a jet source, Cho et al. [39, 27] constructed a thermal source with an oven at 753 K that yielded 1.25×10^{10} molecules per second, of which 2.7×10^{-3} were in the desired $J = 1$ state for an estimated rotational temperature of 350 ± 50 K. The beam velocity is given as 344 ± 20 m/s, with a spread of (150 ± 12) m/s, giving a rotational temperature of 300 ± 50 K. As expected, the beam is both colder and faster than the effusive source from ref. [38]. The higher pressure in a jet source might be expected to result in a lower monomer fraction despite the higher temperature. To counteract this, the authors led the beam from the oven through a 30 cm long nozzle, 20 K hotter than the oven. They calculate that due to the approx. 100 times smaller density of TlF in the nozzle compared to the oven, the monomer fraction is as high as 92%.

The recent trend is towards cryogenic buffer-cooled beam sources. A non-reactive buffer gas is flowed through a copper cell kept at a temperature T_0. The flow rate F of the buffer gas controls the numbre density of the buffer gas in the cell. The species of interest is injected at a temperature much higher than T_0 and thermalizes through with the buffer gas, cooling its rotational and translational degrees of freedom to approximately T_0. Since the cell is at a cryogenic temperatures, all target molecules that collide with the walls stick to it, while others escape through an exit aperture in the cell; the ratio of the time scales of these two processes determines the extraction efficiency of the cell. Beam properties are largely determined by the collisions between the target particles and the buffer gas around the exit aperture. See [40] for a review of cryogenic buffer gas beam sources.

Barry et al. [41] report on a cryogenic buffer-gas source for strontium monofluoride, yielding some 1.2×10^{11} molecules/sr/pulse that are in the desired rovibronic state. At 140 m/s, the beam is slower than a comparable thermal beam, while the rotational temperature is around 1 K. The cell (internal volume 15 cm^3) was kept at 3 K, and the flow of helium controlled at 5 sccm. The beam escapes from the cell through a 3 mm diameter hole in a 0.5 mm thick copper plate. The beam source design has been adapted by Norrgard et al. [42] to produce a beam of thallium fluoride.

In CENTREX, we use neon as the buffer gas. Compared to helium-based sources, neon allows the cell temperature to be higher (∼20 K instead of ∼4 K) for a compable beam temperature after gas expansion. Furthermore, neon is efficiently cryopumped by cold surfaces found in the beam source unlike helium, for which a large-area adsorbent, such as activated charcoal, is necessary. [40]

D. The TlF molecule

TlF is a suitable system to look for P and T violating interactions for a number of reasons: a molecular beam of thallium fluoride can readily be obtained by following the lead of a number of previously published papers; the properties of the molecular states and transitions are available in the literature; the species is a polar diatomic molecule, enhancing the effective electric field acting at the site of the nuclei; the thallium nucleus being very heavy, the Schiff moment is correspondingly relatively large; and the nucleus contains an unpaired proton, making measurements potentially sensitive to proton EDM effects. [33, 31] By contrast, TlF is not a suitable system for electron EDM measurements due to the closed shells zero total electron spin. [43]

Thallium comes in two naturally occurring isotopes, 203 (29.52% abundance) and 205 (70.48%), while ^{19}F is the only isotope of fluorine [44]. In CENTREX, we are only making use of the more abundant isotope 205, and only the transitions between the $v = 0$ vibrational sublevels of the electronic states $X^1\Sigma^+$ and $B^3\Pi_1$, where X denotes the ground state, and B is one of the excited states with a lifetime of about 99 ns [42]. The quantum numbers used to describe the states usually refer to a modified Hund's case (c) basis with the additional couplings $\mathbf{F}_1 = \mathbf{J} + \mathbf{I}_1$ and $\mathbf{F} = \mathbf{F}_1 + \mathbf{I}_2$, where $\mathbf{I}_1$ and $\mathbf{I}_2$ refer to thallium and fluorine nuclear spins, respectively; both nuclei are spin-half.

A particular rotational hyperfine level can be written in either the coupled or decoupled basis, as follows:

$$|\psi_{\text{coup.}}\rangle = |J, \Omega, m_J, I_1, F_1, I_2, F, m_F\rangle$$
$$|\psi_{\text{dec.}}\rangle = |J, \Omega, m_J\rangle|I_1, m_1\rangle|I_2, m_2\rangle,$$

where Ω is the projection of $\mathbf{J}$ on the internuclear axis and m_F is the projection of F in the laboratory frame. [42] The change of basis transformation between the two representations is simply a matter of applying the definition of the Clebsch-Gordan coefficients [45, eqn 5.77] twice to decouple $\mathbf{F}$, then $\mathbf{F}_1$, into the constituent spins $\mathbf{J}$, $\mathbf{I}_1$, and $\mathbf{I}_2$.

Ground state The structure of the ground state can be described by the effective Hamiltonian given in ref. [31]:

$$
\begin{aligned}
\mathcal{H}_{\mathrm{TlF}} &= \mathcal{H}_{\mathrm{rot}} + \mathcal{H}_{\mathrm{S}} + \mathcal{H}_{\mathrm{Z}} + \mathcal{H}_{\mathrm{sr}} + \mathcal{H}_{\mathrm{ss}}, \quad \text{where} \\
\mathcal{H}_{\mathrm{rot}} &= hB\mathbf{J}^2 \\
\mathcal{H}_{\mathrm{S}} &= -\boldsymbol{\mu}_e \cdot \mathbf{E} \\
\mathcal{H}_{\mathrm{Z}} &= -\frac{\mu_J}{J}(\mathbf{J}\cdot\mathbf{B}) - \frac{\mu_1}{I_1}(\mathbf{I}_1\cdot\mathbf{B}) - \frac{\mu_2}{I_2}(\mathbf{I}_2\cdot\mathbf{B}) \\
\mathcal{H}_{\mathrm{sr}} &= c_1(\mathbf{I}_1\cdot\mathbf{J}) + c_2(\mathbf{I}_2\cdot\mathbf{J}) \\
\mathcal{H}_{\mathrm{ss}} &= 5c_3\left(\frac{3(\mathbf{I}_1\cdot\mathbf{J})(\mathbf{I}_2\cdot\mathbf{J}) + 3(\mathbf{I}_2\cdot\mathbf{J})(\mathbf{I}_1\cdot\mathbf{J}) - 2(\mathbf{I}_1\cdot\mathbf{I}_2)\mathbf{J}^2}{(2J+3)(2J-1)}\right) \\
&\quad + c_4(\mathbf{I}_1\cdot\mathbf{I}_2)
\end{aligned}
$$

The energy levels found by diagonalizing this Hamiltonian are illustrated in Fig. 6 in the absence of electric and magnetic fields. The largest effect is the rotational energy $\mathcal{H}_{\mathrm{rot}} = hB\mathbf{J}^2$ with the rotational constant of order $2B = 13.4$ GHz. More precisely, the rotational-vibrational energy levels of a diatomic molecule are given by the so-called Dunham expansion [46]

$$T_{vJ} = \sum_{lm} Y_{lm}(v+1/2)^l[J(J+1)]^m.$$

In practice, however, we find that the lowest few microwave rotational transitions are sufficiently well described using only the rotational constant in equilibrium position $B_{\mathrm{e}} \equiv Y_{01}$ and the first term $\alpha_{\mathrm{e}} \equiv Y_{11}$ using NIST data [47], yielding the effective constant $2B = 13.28958$ GHz for the isotope $^{205}\mathrm{Tl}^{19}\mathrm{F}$. Note that the value $2B = 13.37984$ GHz, given in Ramsey et al. [31], only takes into account B_{e} and is thus not appropriate for driving microwave transitions.

Due to a coupling of molecular rotation to the nuclear spin magnetic moments, hyperfine levels are made non-degenerate by an interaction $\mathcal{H}_{\mathrm{sr}} = c_1(\mathbf{I}_1\cdot\mathbf{J}) + c_2(\mathbf{I}_2\cdot\mathbf{J})$, with $c_1 \approx 126$ kHz and $c_2 \approx 18$ kHz. The next largest effect, the spin–spin interaction, is due to two distinct effects of magnitudes $c_3 = 0.070$ kHz and $c_4 = -13.30$ kHz. The c_3 term is due to a direct dipole–dipole coupling between the nuclear spins. In the Hamiltonian given above, the c_3 term is written in an effective form, valid for small (less than approx. 20 kV/cm) electric fields when J is a good quantum number. At large E-field values, J is no longer a good quantum number, and this approximation cannot be used. In its stead, Brown and Carrington [45, eqn 4.17] give

$$\mathcal{H}_{\mathrm{ddd}} = \frac{g_1 g_2 \mu_N^2}{4\pi\varepsilon_0 c^2}\left\{\frac{\mathbf{I}_1\cdot\mathbf{I}_2}{R^3} - \frac{3(\mathbf{I}_1\cdot\mathbf{R})(\mathbf{I}_2\cdot\mathbf{R})}{R^5}\right\}, \tag{34}$$

where $g_{1,2}$ are the nuclear g-factors and $\mathbf{R}$ is the internuclear displacement vector. The interaction is proportional to a spherical rank-2 tensor $T^2(\mathbf{I}_1, \mathbf{I}_2)$ as per eqn 1.34 in ref. [45].

At first glance, the more general form of the c_3 term does not change the eigenenergies significantly. However, the slope of the NMR resonance frequency as a function of the applied field is a factor of ≈ 2 greater when using eqn 34, as will be seen in Fig. 9, reproducing the result 41 mHz/(V/cm) reported at 20 kV/cm in Ramsey et al. [31], while the form of the Hamiltonian as given in that paper gives only 20.2 mHz/(V/cm). The later Hinds et al. paper [27, 39] seems to have copied the slope from Ramsey, claiming it holds at 29.5 kV/cm, where the correct slope (as per eqn 34) would be 31.5 mHz/(V/cm).

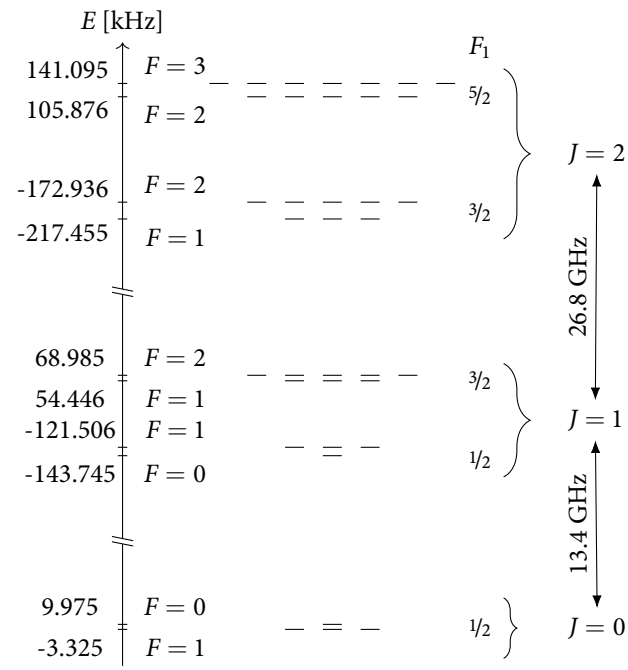

Figure 6: First three rotational levels in the TlF ground state $X^1\Sigma^+$ zero-field energy levels. Rotational energies $BJ(J+1)$ are subtracted from the energies indicated on the axis. Note the $(2F+1)$-fold degeneracy, corresponding to the quantum numbers $m_F = -F, \ldots, 0, \ldots, F$. The numbers left of the braces are the F_1 labels.

The c_4 term, also known as the scalar- or J-coupling term, arises through a mechanism involving the transmission of nuclear spin orientation through the interven- ing electrons [45]. It is distinguished from the tensor interaction in that it does not depend on the angle between the nuclear directions and the internuclear axis $\mathbf{R}$. While molecular rotation averages the c_3 term to zero, the c_4 term survives [48, sect. 7.7]. In spherical tensor form, the interaction is proportional to $T^1(\mathbf{I}_1)\cdot T^1(\mathbf{I}_2)$.

For most of the terms in the effective Hamiltonian, the matrix elements are most readily calculated in the decoupled basis, where the operators $\mathbf{J}^2$, J_z, I_{1z}, I_{2z} are diagonal. Most of the other operators can be written in terms of the appropriate ladder operators $J_\pm$ [49, eqns 3.5.39–40]

$$J_\pm|J,m\rangle = \sqrt{(J\mp m)(j\pm m+1)}|J,m\pm 1\rangle, \tag{35}$$

and similarly for $I_{1\pm}$, $I_{2\pm}$ acting on the $|I_1, m_1\rangle$ and $|I_2, m_2\rangle$ subspaces. For instance, $J_x = (J_+ + J_-)/2$ and $J_y = (J_+ - J_-)/2i$. The dot product of two angular momentum operators can be written in terms of the ladder operators as

$$\mathbf{A}\cdot\mathbf{B} = A_z B_z + \frac{1}{2}\left(A_+B_- + A_-B_+\right). \tag{36}$$

Equations 35 and 36, together with the fact the basis states with dissimilar labels are orthogonal, allow us to find the matrix elements between arbitrary basis states for all terms in the Hamiltonian except for the Start effect, to which we turn now.

The Stark Hamiltonian can be expressed explicitly in the components of the molecular electric dipole moment and the lab-frame electric field:

$$\mathcal{H}_s = \mu_e \sqrt{\frac{2\pi}{3}} \begin{pmatrix} Y_1^{-1} - Y_1^1 \\ i(Y_1^{-1} + Y_1^1) \\ \sqrt{2} Y_1^0 \end{pmatrix} \cdot \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix},$$

with the matrix elements written in terms of the Clebsch-Gordan coefficients $\langle j_1 m_1 j_2 m_2 | j m \rangle$

$$\langle J', m_J' | Y_1^M || J, m_J \rangle = \sqrt{\frac{3}{2\pi}} W_M, \quad \text{where}$$

$$W_M = (-1)^M \sqrt{\frac{2J'+1}{2(2J+1)}} \langle J' \, 0 \, 1 \, 0 | J \, 0 \rangle \langle J' \, m_J' \, 1 \; -M | J \, m_J \rangle.$$

For $M = 0$, the expression is nonzero only for $m_J' = m_J$:

$$W_0 = \begin{cases} \sqrt{\dfrac{(J-m_J)(J+m_J)}{8J^2-2}} & \text{for } J' = J-1 \\ \sqrt{\dfrac{(J-m_J+1)(J+m_J+1)}{6+8J(J+2)}} & \text{for } J' = J+1 \end{cases}$$

For $M = \pm 1$, we need $m_J' = m_J \pm 1$:

$$W_{\pm 1} = \begin{cases} -\dfrac{1}{2}\sqrt{\dfrac{(J \mp m_J)(J-1 \mp m_J)}{4J^2-1}} & \text{for } J' = J-1 \\ \dfrac{1}{2}\sqrt{\dfrac{(J+1 \pm m_J)(J+2 \pm m_J)}{3+4J(J+2)}} & \text{for } J' = J+1 \end{cases}$$

Since the direction of motion of electrons in the atom does not change the electric dipole moment, time reversal is a symmetry of the Stark effect. [50] It can be shown that there exist pairs of states of opposite m_J with identical matrix elements of the Stark operator S_z (identical polarizabilities to first order). [51] Considering the $M = 0$ expression above, corresponding to the z-direction of the dipole moment, we see the $\mathrm{sgn}(m_J)$ dependence is gone.

We can make use of the expressions for the Stark effect matrix elements in two other contexts. To calculate the degree of molecular polarization (eqn 30), all it takes is the find the expectation value of the z-component of the Stark Hamiltonian in an eigenstate at particular field value, divided by $\mu_e =$ 4.2 Debye. The other application is to find the matrix element of the alternative c_3 term (eqn 34), where the dot products $\mathbf{I}_i \cdot \mathbf{R}$ can be found in spherical tensor notation as

$$\mathbf{I}_i \cdot \mathbf{R} = \sqrt{2} I_{i,0} W_0 + (I_{i,1} W_{-1} - I_{i,-1} W_1).$$

Excited state While the nuclear magnetic resonance occurs entirely within the $X^1\Sigma^+$ ground state, the structure of the excited state $B^3\Pi_1$ is nevertheless important for the state preparation and readout, which we implement using $X \leftrightarrow B$ transitions. In the following, I will present Ω-doubling in the excited state, transition nomenclature, and optical cycling.

States $|J, \Omega, m_J, I_1, F_1, I_2, F, m_F\rangle$ are not eigenstates of parity. The appropriate combinations of these states with a definite parity can be written as the Ω-doublets

$$|\Omega, \pm\rangle = \frac{1}{\sqrt{2}} \left[|\Omega\rangle \pm (-1)^{J+1} | -\Omega\rangle \right], \tag{37}$$

with all the other quantum numbers being implied. (Note that this is not necessary in the ground state, where $\Omega = 0$.) Thus for each J, there is a pair of nearly degenerate states of opposite values of parity $\pm(-1)^{J+1}$, but the ordering of these states alternates with J. To avoid the alternation, integer-J states with parity $-(-1)^{J+1}$ are conventionally designated e-states, and those with $(-1)^{J+1}$ are called f-states. [45, sect. 6.9.4]

According to accepted terminology, a transition is said to belong to the P, Q, or R branches if the quantum number J changes by -1, 0, or $+1$, respectively. [52, sect. 10.2] Furthermore, we append the J number of the original state, and the F of the target state. For instance, a $P2\,F1$ line starts in a $J = 2$ state, changes J by -1, and ends up in a state with $F' = 1$.

Despite the limited ability to collect and detect photons from fluorescing molecules, our detection scheme could in principle detect $\approx$ 100% of the molecules through optical cycling. A laser excites the $X \to B$ transition, causing photons in the decay back down to X. If the molecule ended up in the same sublevel in X, the same laser can re-excite it back to B. [53]

However, due to the large number of degrees of freedom in molecules, optical cycling can be difficult to achieve in molecules. It turns out that in TlF, there exists sufficiently closed transitions to allow $\sim$ 100 cycles before the molecule gets lost in inaccessible dark states (see ref. [42] for some possible candidates). The number of dark states is potentially very large since the hyperfine structure (h.f.s.) of the ground state is unresolved.

Branching ratios The strength of the transitions between various hyperfine states is a function of the matrix elements of the dipole operator $\mathbf{d}$ between the states, which we now proceed to evaluate. Working in the basis $|J, \Omega, m_J, I_1, F_1, I_2, F, m_F\rangle$, with $I_1 = I_2 = 1/2$, use the Wigner-Eckart theorem [45, eqn 5.172]

$$\langle j, m | \mathrm{T}_p^k(\mathbf{A}) | j', m' \rangle = (-1)^{j-m} \begin{pmatrix} j & k & j' \\ -m & p & m' \end{pmatrix} \langle j || \mathrm{T}^k(\mathbf{A}) || j' \rangle \tag{38}$$

to take the m_F dependence out of the matrix element:

$$\begin{aligned} &\langle F, m_F, \ldots | \mathrm{T}_p^1(\mathbf{d}) | F', m_F', \ldots \rangle \\ &\quad = (-1)^{F-m_F} \begin{pmatrix} F & 1 & F' \\ -m_F & p & m_F' \end{pmatrix} \langle F, \ldots || \mathrm{T}^1(\mathbf{d}) || F', \ldots \rangle, \end{aligned} \tag{39}$$

Space and spin commute ($[\mathbf{r}, \mathbf{I}_1] = [\mathbf{r}, \mathbf{I}_2] = 0$), whereas $[\mathbf{r}, \mathbf{J}] \neq 0$. Thus, $\mathbf{I}_1, \mathbf{I}_2$ are 'spectator' spins, i.e., the dipole operator d_p acts only on a part of the coupled scheme. In this case, we can apply the 'spectator theorem' [45, eqn 5.174]

$$\begin{aligned}\langle j_1, j_2, j || \mathrm{T}^{k_1}(\mathbf{A}_1) || j_1', j_2', j' \rangle &= \delta_{j_2, j_2'} (-1)^{j'+j_1+k_1+j_2} \\ &\sqrt{(2j+1)(2j'+1)} \begin{Bmatrix} j_1' & j' & j_2 \\ j & j_1 & k_1 \end{Bmatrix} \langle j_1 || \mathrm{T}^{k_1}(\mathbf{A}_1) || j_1' \rangle,\end{aligned} \tag{40}$$

so that j_2 drops out of the reduced matrix element. (Note: do not use eqn 5.136 in [45] due to a typo in the $6j$-symbol.) We can use that to break up the $\mathbf{F} = \mathbf{F}_1 + \mathbf{I}_2$ coupling:

$$\begin{aligned}\langle F, F_1, I_2 \ldots || \mathrm{T}^1(\mathbf{d}) || F', F_1', I_2', \ldots \rangle &= (-1)^{F'+F_1+1+1/2} \\ &\sqrt{(2F+1)(2F'+1)} \begin{Bmatrix} F_1' & F' & 1/2 \\ F & F_1 & 1 \end{Bmatrix} \langle F_1, \ldots || \mathrm{T}^1(\mathbf{d}) || F_1', \ldots \rangle.\end{aligned} \tag{41}$$

We can repeat this procedure on $\mathbf{F}_1 = \mathbf{J} + \mathbf{I}_1$, taking $\mathbf{I}_1$ to be the spectator spin:

$$\begin{aligned}\langle F_1, J, I_1 \ldots || \mathrm{T}^1(\mathbf{d}) || F_1', J', I_1', \ldots \rangle &= (-1)^{F_1'+J+1+1/2} \\ &\sqrt{(2F_1+1)(2F_1'+1)} \begin{Bmatrix} J' & F_1' & 1/2 \\ F_1 & J & 1 \end{Bmatrix} \langle J, \ldots || \mathrm{T}^1(\mathbf{d}) || J', \ldots \rangle.\end{aligned} \tag{42}$$

In order to have an expression for the non-reduced matrix element, we can use Wigner-Eckart in reverse:

$$\langle J, \ldots || \mathrm{T}^1(\mathbf{d}) || J', \ldots \rangle = \frac{\langle J, m_J, \ldots | \mathrm{T}^1_p(\mathbf{d}) | J', m_J', \ldots \rangle}{(-1)^{J-m_J} \begin{pmatrix} J & 1 & J' \\ -m_J & p & m_J' \end{pmatrix}} \tag{43}$$

Note that the dipole operator $\mathbf{d} = -e\mathbf{r}$ is defined in the molecule-fixed frame. To transform from the lab frame (indices p) to the molecule frame (indices q), we can use the relation [45, eqn 5.182]

$$\mathrm{T}^k_p(\mathbf{A}) = \sum_q \mathcal{D}^{(k)}_{pq}(\omega)^* \mathrm{T}^k_q(\mathbf{A}),$$

where $\mathcal{D}^{(k)}_{pq}$ are the Wigner rotation matrices as a function of angular coordinates ω. We can use this to rewrite the last matrix element above as follows:

$$\begin{aligned}&\langle J, m_J, \ldots | \mathrm{T}^1_p(\mathbf{d}) | J', m_J', \ldots \rangle \\ &= \langle J, m_J, \ldots | \sum_q \mathcal{D}^{(1)}_{pq} \mathrm{T}^1_p(\mathbf{d}) | J', m_J', \ldots \rangle\end{aligned}$$

The operator $\mathrm{T}^1_q(\mathbf{d})$ acts in molecule-fixed frame, in which there is no molecular rotation, so it cannot act on the mechanical rotation part of the J angular momentum. In does, however, act on the $\mathbf{\Omega}$ part, allowing us to split the above into

$$\sum_q \langle J, m_J, \Omega | \mathcal{D}^{(1)}_{pq} | J', m_J', \Omega' \rangle \underbrace{\langle \Omega | \mathrm{T}^1_q(\mathbf{d}) | \Omega' \rangle}_{\delta_{q,\Omega-\Omega'} \langle \mathbf{d} \rangle}.$$

The second factor, including the $\delta_{q,\Omega-\Omega'}$, in the expression is the same for all X–B transitions. For the first one, we can use the formula [45, eqn 5.184]

$$\begin{aligned}\langle J, \Omega, m | \mathcal{D}^{(k)}_{pq}(\omega)^* | J', \Omega', m' \rangle &= (-1)^{m-\Omega} \\ &\sqrt{(2J+1)(2J'+1)} \begin{pmatrix} J & k & J' \\ -m & p & m' \end{pmatrix} \begin{pmatrix} J & k & J' \\ -\Omega & q & \Omega' \end{pmatrix}\end{aligned}$$

with q set to $\Omega - \Omega'$:

$$\begin{aligned}\langle J, m_J, \Omega | \mathrm{T}^1_p(\mathbf{d}) | J', m_J', \Omega' \rangle &= \langle \mathbf{d} \rangle (-1)^{m_J-\Omega} \\ &\sqrt{(2J+1)(2J'+1)} \begin{pmatrix} J & 1 & J' \\ -m_J & p & m_J' \end{pmatrix} \begin{pmatrix} J & 1 & J' \\ -\Omega & \Omega-\Omega' & \Omega' \end{pmatrix}.\end{aligned} \tag{44}$$

Finally, it remains to collect together all the factors from equations 39, 41–44. Rearranged for readability, we have:

$$\begin{aligned}&\langle F, m_F, \ldots | \mathrm{T}^1_p(\mathbf{d}) | F', m_F', \ldots \rangle = \\ &\quad \langle \mathbf{d} \rangle (-1)^{3+F-\Omega+F_1-m_F+F'+F_1'} \big[(2J+1)(2J'+1) \\ &\quad (2F_1+1)(2F_1'+1)(2F+1)(2F'+1) \big]^{1/2} \\ &\quad \begin{Bmatrix} J' & F_1' & 1/2 \\ F_1 & J & 1 \end{Bmatrix} \begin{Bmatrix} F_1' & F' & 1/2 \\ F & F_1 & 1 \end{Bmatrix} \\ &\quad \begin{pmatrix} J & 1 & J' \\ -\Omega & \Omega-\Omega' & \Omega' \end{pmatrix} \begin{pmatrix} F & 1 & F' \\ -m_F & p & m_F' \end{pmatrix}.\end{aligned} \tag{45}$$

As mentioned above eqn 37, the decoupled basis states are not parity eigenstates in the excited states. Thus, the appropriate matrix elements to use are the symmetrized combinations

$$\frac{1}{\sqrt{2}} \big[\langle \Omega | \mathrm{T}^1_p(\mathbf{d}) | \Omega' \rangle \pm (-1)^{J+1} \langle \Omega | \mathrm{T}^1_p(\mathbf{d}) | -\Omega' \rangle \big], \tag{46}$$

emphasizing again that the doublets only occur in the excited state, where $\Omega \neq 0$. Furthermore, in the excited state, energy eigenstates contain an admixture of multiple different $|J, F_1, F\rangle$ basis states as detailed in Table IV of ref. [42]; these must be taken into account for an accurate calculation of physical matrix elements.

Appropriately normalized, eqns 45 and 46 yield the branching ratios $b_{\mathrm{e}\to\mathrm{g}}$ from a given sublevel $|\mathrm{e}\rangle$ in the excited state to a sublevel $|\mathrm{g}\rangle$ in the ground state. The squared magnitude of the matrix element needs to be divided by the sum over all possible final (ground) states from the given initial (excited) state:

$$b_{\mathrm{e}\to\mathrm{g}} \equiv \frac{\sum_p |\langle \mathrm{e} | \mathrm{T}^1_p(\mathbf{d}) | \mathrm{g} \rangle|^2}{\sum_{|\mathrm{g}\rangle} \sum_p |\langle \mathrm{e} | \mathrm{T}^1_p(\mathbf{d}) | \mathrm{g} \rangle|^2}, \tag{47}$$

explicitly writing out the sum over polarizations p. This allows us to immediately calculate branching ratios between pure-parity states given the quantum numbers of each state; see Appendix B in ref. [54] for explicit numbers.

E. Previous experiments

The first P, T-violation experiment using a molecular beam of thallium fluoride was done in Oxford in 1969 [55], the result being reported as $\nu = 0.08 \pm 0.11$ Hz, or, in terms of proton EDM, as $|d_{\mathrm{p}}| = (7 \pm 9) \times 10^{-21} e$ cm. They employed the triple resonance scheme: first, $J = 1$ molecules were transferred from $m_J = 0$ to $m_J = \pm 1$. The second resonance which changed the Tl nuclear spin projection from $\pm 1/2$ to $\mp 1/2$ is not directly observable; however, after the molecules are transferred back to $m_J = 0$, the amount of the non-observable spin flips can be estimated and related to the desired frequency shifts 4ν.

About a decade later, Hinds and Sandars [30] reported a similar experiment giving the null result of $\nu = 11 \pm 42$ mHz or $|d_{\mathrm{p}}| = (1.8 \pm 7) \times 10^{-21} e$ cm. Looking for P and T violating interactions in ^{205}TlF, they tried to measure shifts of the resonant frequency which depends on the relative directions of externally applied parallel electric and magnetic fields. The experiment used a beam of TlF produced by vapors effusing from a hot oven through a state-selecting region that approximately transmits only molecules with $J = 1$, $m_J = -1$, $m_{\mathrm{Tl}} = +1/2$, $m_{\mathrm{F}} = -1/2$. In the interaction region, separated oscillating magnetic fields induce some molecules to flip the Tl spin to $m_{\mathrm{Tl}} = -1/2$. Finally, a second state selector transmits only the molecules without the spin flip, and the final beam intensity, as measured by a hot-wire surface ionization detector, is a sensitive measure of the shift in the resonance given that the 10 m separation of the oscillating coils results in a 20 Hz linewidth. However, the experiment gave an accuracy comparable to the experiment in [55] because of an unexpectedly large reduction of the resonance signal and high background rate.

Using a similar setup, Wilkening, Ramsey, and Larson [31] reduced the error bars to $\nu = 8 \pm 21$ mHz or $|d_{\mathrm{p}}| = (1.3 \pm 2.0) \times 10^{-21} e$ cm. In their experiment, the thermal beam of TlF passes through an electrostatic quadrupole field that focuses the $J = 1$, $m_J = 0$ state, leaving the others unselected. Thereafter, a 'subsidiary resonance' region transfers the molecules into the state $m_J = -1$, $m_{\mathrm{Tl}} = -1/2$, $m_{\mathrm{F}} = -1/2$. However, instead of using oscillating fields to do so, it relies on adiabatic transitions that conveniently happen to transfer population into the desired state. As was the case in the preceding experiment, the main interaction region causes the Tl nuclear spin to flip. Finally, second subsidiary resonance region transfers the flipped molecules to the $m_J = -1$ hyperfine state which is defocused by the second quadrupole, causing a change in the beam intensity measured by the hot-wire detector at the end of the beamline.

Cho, Sangster, and Hinds [32, 27] found the upper bound $\nu = (1.4 \pm 2.4) \times 10^{-4}$ Hz or $|d_{\mathrm{p}}| = (-3.7 \pm 6.3) \times 10^{-23} e$ cm. Again, a quadrupole selects $J = 1$, $m_J = 0$ molecules, which subsequently transition to one of the $m_J = \pm 1$ sublevels in a combination of static and RF electric fields, with quantum numbers $(m_J, m_{\mathrm{Tl}}, m_{\mathrm{F}}) = (-1, +1/2, -1/2)$ or $(+1, -1/2, +1/2)$. Then, separated oscillating magnetic fields in the presence of a strong electric field induce a nuclear spin flip in Tl, observed after a second state selector and quadrupole filter by measuring the ion current produced by a hot, oxygenated tungsten filament. The improved upper bound was due to a more intense and rotationally colder beam of TlF monomers, and due to an additional reversal, switching between the two hyperfine states as listed above.

Other systems While the thallium fluoride experiments look for proton EDMor the Schiff moment, other experiments search for electron or neutron EDMs, or atomic EDMs. There has been a string of experiments on atomic thallium. Already in the 1970s, an experiment reported $d_{\mathrm{Tl}} = (1.3 \pm 2.4) \times 10^{-21} e$ cm looking for a shift between the $F = 1$, $m_F = 0, 1$ sublevels of the $6^2P_{1/2}$ ground state of Tl in a uniform electric field of 50 kV/cm in the presence of a 1 G magnetic field parallel to the electric field. [56] In 1994, E. Commins et al. reported $d_e = (1.8 \pm 1.2 \pm 1.0) \times 10^{-26} e$ cm, referring to the statistical and systematic uncertainties, respectively. [57] The experiment was subsequently significantly upgraded, improving the precision to $|d_e| < 1.6 \times 10^{-27} e$ cm. [58]

A measurement on ultracold neutrons stored in a trap gave the result $|d_{\mathrm{n}}| < 2.9 \times 10^{-26} e$ cm. Since the particle were kept in a trap, the 130 s precession time between the (temporally) separated oscillatory fields is significantly longer than is commonly found in beam-based measurements. [10]

In a decades-long search for the EDM of ^{199}Hg, the Zeeman precession frequency of the nuclear spins in parallel electric and magnetic fields was measured. To reduce the frequency noise due to magnetic field fluctuations, two mercury cells are positioned next to each other but with opposite electric field direction, analogously to the electric field reversals typical of molecular-beam experiments. [59]. The result has been constrained to $|d_{\mathrm{Hg}}| < 7.4 \times 10^{-30} e$ cm, which can also be interpreted in terms of a neutron EDM limit $|d_{\mathrm{n}}| < 1.6 \times 10^{-26} e$ cm, superseding ref. [60] as the best such limit. [61, 9]

F. Overview of the CENTREX scheme

Referring to Fig. 7, we follow the molecules emerging from a cryogenic buffer gas beam source, initially in a range of low-lying states with an approximate temperature of 8 K. In order to facilitate transfer to the 'science' state used for the spin precession measurement, molecules first enter a rotational cooling region to accumulate population in a single hyperfine and rotational level. Then, to limit the spreading of the beam, the molecules get focused by an quadrupole electrostatic lens. Since the lens state is different from the interaction state, the molecules are prepared into the correct hyperfine state in a two-state adiabatic rapid passage machine. Afterwards, they fly through an interaction region with the standard Ramsey separated oscillatory field configuration to excite the NMR resonance. Finally, we probe the resulting spin direction to determine the extent

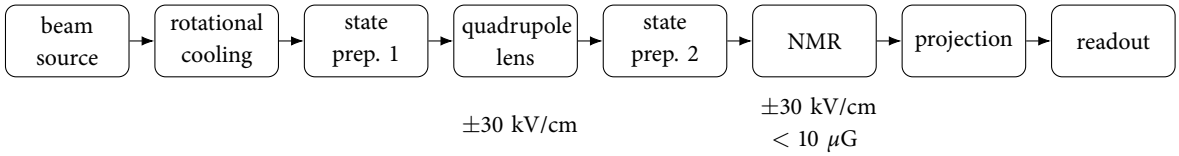

Figure 7: Conceptual flow chart of the critical steps in CENTREX, indicating the applied fields (below) and the quantum states (above).

of the possible dependence of the spin precession frequency on the interaction region electric field direction.

Rotational cooling (Fig. 8) Assuming a simple Boltzmann distribution at 8 K, almost 90% of the molecules have J in the range from 0 to 3. To move these molecules to $J = 0$, we first set up microwave couplings $J = 1 \leftrightarrow 2$ and $2 \leftrightarrow 3$. Then, a UV laser beam centered on the $P2$ $F1$ line lifts the population from $J = 2$ into the excited state $J' = 1$, from where it decays down to either $J = 0$ or $J = 2$. Since the $J = 2$ molecules eventually get excited by the laser again, we expect the population to accumulate in $J = 0$.

The preceding description ignores the complications due to the hyperfine structure. Since the excited state hyperfine levels are well separated (unlike the unresolved h.f.s.of the electronic ground state), we can address the $F' = 1$ level. Due to the selection rule $\Delta F = 0, \pm 1$, only three of the four $J = 2$ sublevels (with $F = 1, 2, 2$) can be excited into $J' = 1$, from where they fall down into $J = 0$ states with $F = 1, 0$. A further set of microwaves, between $J = 0$ and $J = 1$, is used to remix the $F = 1$ states, so as to concentrate the population only in the $F = 0$ state. (If we were to use the $P2$ $F2$ transition instead, then it would be possible to excite all four F sublevels of the $J = 2$ state up into a $F' = 2$ level. However, the branching ratio—calculated using eqn 47—for decays to $F = 0$ are vanishing from $F' = 2$.)

As stated, the $P2$ $F1$ does not excite the $F = 3$ states, so these are trivially dark. But since there are only 3 sublevels in $F' = 1$, compared to the 20 sublevels in the $J = 2$ manifold, there is a total of 17 dark states. These may be destabilized by frequency-switching microwaves precisely tuned to the individual sublevels that cannot be resolved by the laser light; however the precise dark-state destabilization scheme has not yet been finalized.

State transfer 1 The population that has been collected in the $J = 0$, $F = 0$ sublevel has to be moved to the state in the $J = 2$ manifold chosen for the optimal operation of the electrostatic lens. We plan to accomplish the transfer as a two-stage adiabatic rapid passage (ARP): first $J = 0 \rightarrow 1$, then $1 \rightarrow 2$. As the particles travel through a spatially-varying electric field, the separation between the hyperfine levels in the different J states changes due to the DC Zeeman effect. In the same region, we place two microwaves beams corresponding to each pair of J states, and offset them by approximately the FWHM of each beam. As the molecule traverses each beam, it is successively driven through the respective microwave resonances, making the population jump first to $J = 1$, and then to $J = 2$.

To keep the state transfer adiabatic, the electric field (and the resulting level separation, and the detuning $\Delta\omega(t)$ between the microwaves and the state spacing) has to change slowly compared with the Rabi frequency of the microwaves $\Omega_{\mu w}$. For a Rabi frequency much greater than the detuning, ref. [62] gives the adiabaticity condition as

$$\Omega_{\mu w} \gg \frac{d}{dt}\Delta\omega(t).$$

The other requirement for ARP is that the passage is *rapid* compared to the spontaneous lifetime of the states involved in the transition, which here is effectively infinite. [63]

Electrostatic lens Molecules leave the beam source spread over a wide solid angle, so the beam intensity decreases roughly as the distance squared from the source. To counteract this effect, we intend to collimate the beam, or focus it on the detection region, some seven meters away from the source.

It turns that that an appropriate field distribution can be used as an analogue to the optical lens [64], the condition being that inside the lens, molecules experience a restoring force proportional to the displacement r of the molecule from the axis. For a quadrupole field, the magnitude of the field is $|E| = Vr/R^2$, where V is the potential difference between oppositely-charged electrodes and $2R$ the distance between opposite electrodes. [65] Furthermore, since the $J = 2$ state of thallium fluoride has a quadratic Stark effect at high fields ($\propto 30$ kV/cm), this results in the desired quadratic potential energy, or a linear restoring force. Ref. [66] makes the optical analogy exact in terms of ray-transfer (or 'ABCD') matrices, whereby the condition for focusing results in the standard expression for a thin lens of focal length f

$$\begin{pmatrix} r \\ \dot{r}/u \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -1/f & 1 \end{pmatrix} \begin{pmatrix} r_0 \\ \dot{r}_0/u \end{pmatrix},$$

where u is the component of velocity along the beam direction (making $\dot{r}_0/u$ proportional to the angle relative to the axis), and

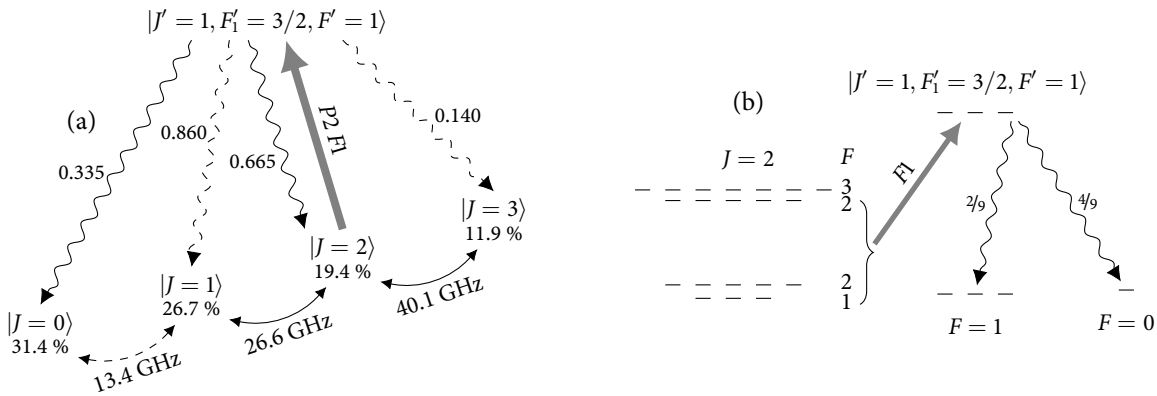

Figure 8: Proposed rotational cooling scheme. ***a.*** The thick solid arrow marks the UV laser driving the $P2$ $F1$ transition, while the bent arrows represent microwaves with the stated frequencies (the dashed 13.4 GHz microwaves might be omitted in the final version). The wavy arrows indicate spontaneous emission, with rotational branching ratios taken from ref. [42, Table IV]. The odd-parity Ω-doublet excited state can only decay to the states with $J = 0, 2$ (solid wavy lines), and not the $J = 1, 3$ ones (dashed wavy lines), and the branching ratios are normalized to unity for each parity. The percentages under the ground-state kets are the expected thermal population at 8 K before beginning rotational cooling. ***b.*** Hyperfine details of the rotational cooling scheme. Decays back to $J = 2$ not shown here. The $P2$ $F1$ transition, shown here, does not couple to $F = 3$. However, the $P2$ $F2$ line (not shown) that does couple to all F has a vanishing branching ratio into $F = 0$.

$r_0(t)$ the pre-lens displacement of the molecule from the beam axis. The focal length for molecules of mass m is [66, cf. eqn 12]

$$f = \frac{1}{p \sin pl}, \quad p = -\frac{CV^2}{R^4 m u^2},$$

where l is the length of the quadrupole lens, and C is a constant encoding the J, m_J dependence of the quadratic Stark effect.

State transfer 2 A second state preparation region will take molecules out of the lens $J = 2$ state and into the appropriate $J = 1$ state required by the interaction region. Details will be decided upon based on experimental results from the first state preparation region.

Interaction region A high external electric field polarizes the molecules, thus enabling the linear energy shift due to the Schiff moment. The measurement will be done using the method of separated oscillatory fields as explained above.

We intend to use a pair of $J = 1$, $m_J = \pm 1$ states, where the molecular rotation leads to an internal magnetic field. By flipping the sign of m_J, the internal magnetic field can be reversed. The first RF coil in the Ramsey sequence creates a superposition of thallium spin-up and spin-down states, and the phase accumulated during the interaction time is translated by the second pulse back into varying populations of the spin states. The data obtained with the two different fluorine spin directions can be used for systematic checks.

Refer to Table I for a list of all available pairs of states we can make use of. For the EDM measurement, of course, only the transitions that flip the Tl nuclear spin are suitable. We quantify the sensitivity S_{SMt} to the T-violating physics as the extent to which the transition between states i and f flips the thallium spin:

$$S_{\text{SMt}} \equiv |\langle I_{1,z}\rangle_i - \langle I_{1,z}\rangle_f| \tag{48}$$

We can see that this quantity has a significant magnitude only for the three pairs of transitions that are labelled as flipping m_{Tl}. Of these, the first pair listed (ej/hk) is the one used in the papers by Ramsey et al. [31] and Hinds et al. [27] One pair of transitions (ef/hg) has a much smaller resonant frequency than the other two, but is otherwise comparable.

The pair fk/gj with $\Delta m_J = 0$ is significantly less sensitive to the residual magnetic field in the interaction region, and to imperfections in the electric field reversals, but requires the applied RF magnetic field to be parallel with the electric field unlike all the other two transitions. That pair of transitions also flips the fluorine spin, but in the opposite sense to the change of m_{Tl}. The sensitivity to external fields is greatly reduced since the magnetic dipole moments of these nuclei are relatively well matched [67]:

$$\mu({}^{205}_{81}\text{Tl}) = +1.63821461(12)\mu_{\text{N}}$$
$$\mu({}^{19}_{9}\text{F}) = +2.628868(8)\mu_{\text{N}}.$$

@Dave: They are not really all that well-matched! How do you then explain the cancellation of sensitivity to the external magnetic field? And why are these states less sensitive to E-reversal imperfections?

Of the three pairs of transitions that do not flip m_{Tl}, the pair fj/gk is exactly forbidden. The other two (ek/hj and eg/hf) are about as sensitive to field imperfections as the two pairs of m_{Tl}-flipping transitions ej/kh and ef/hg, but $\approx$ 23 times less sensitive to the T-violating physics. These states can thus be used for magnetometer/electrometer transitions in calibrating the experimental systematics.

Table I: All non-degenerate pairs of $|m_J| = 1$ states in the asymptotic $J = 1$ manifold that do not involve the states $|\mathrm{i}\rangle$ and $|\mathrm{l}\rangle$. The quantum numbers given are that of the largest decoupled-basis component; states i and l have two such components with equal amplitude, making it hard to say what spin is flipped. The labels l_R are as given in ref. [31], while the indices i refer to the ordering of eigenvalues at 30 kV/cm, starting with 0–3, the lowest four $J = 0$ states. The column labelled 'S_SMT' gives the sensitivity to the Schiff moment (eqn 48). The quantity S_m gives the energy difference between the neighbouring resonances, given per unit external (residual) magnetic field, and corresponds to flipping the internal molecular magnetic field. Due to degeneracy of pairs (e, h), (f, g), and (j, k), there are only three distinct resonant frequencies f_0 and slopes df_0/dE_z at 30 kV/cm.

	State 1					State 2					$\lvert\langle\cdot\lvert\mathcal{H}_\mathrm{Z}\rvert\cdot\rangle\rvert$ [kHz]		S_SMT	S_m	f_0	df_0/dE_z
What flips?	l_R	i	m_J	m_Tl	m_F	l_R	i	m_J	m_Tl	m_F	x, y	z	—	mHz/μG	kHz	mHz/(V/cm)
m_Tl	e	10	−	−	−	j	7	−	+	−	1.33	0.00	0.95	−8.09	119.517	−31.50
	h	11	+	+	+	k	6	+	−	+						
m_J, m_F	e	10	−	−	−	k	6	+	−	+	0.28	0.00	0.04	−4.99		
	h	11	+	+	+	j	7	−	+	−						
m_J	f	8	+	+	−	j	7	−	+	−	0.00	0.00	0.00	−3.01	108.924	−3.57
	g	9	−	−	+	k	6	+	−	+						
$m_\mathrm{Tl}, m_\mathrm{F}$	f	8	+	+	−	k	6	+	−	+	0.00	0.09	0.99	+0.09		
	g	9	−	−	+	j	7	−	+	−						
m_F	e	10	−	−	−	g	9	−	−	+	1.88	0.00	0.04	−8.00	10.593	−27.93
	h	11	+	+	+	f	8	+	+	−						
m_J, m_Tl	e	10	−	−	−	f	8	+	+	−	0.40	0.00	0.95	−5.07		
	h	11	+	+	+	g	9	−	−	+						

There are a further 13 possible transitions, not listed in Table I, involving the 'superposition' states $|\mathrm{i}\rangle$ and $|\mathrm{l}\rangle$, given in [27, Table I] with the following $|m_J, m_1, m_2\rangle$ eigenvectors:

$$|\mathrm{i}\rangle, |\mathrm{l}\rangle = \frac{1}{\sqrt{2}}\big(\,|+,-,-\rangle \pm |-,+,+\rangle\,\big).$$

These states can be ignored, since only one ket of the pair flips the thallium spin when transitioning to a non-superposition state, and as such can be only half as sensitive to T-violating physics. On the other hand, the transition between $|\mathrm{i}\rangle$ and $|\mathrm{l}\rangle$ does not flip m_Tl at all.

Projection and detection While the detection scheme has not yet been finalized, we expect we will be able to map the two spin states from the interaction region to two different rotational states using microwaves; details TBD. Unlike some of the experiments discussed above, CENTREX will employ laser-induced fluorescence (LIF) to detect the number of molecules in the final rotational states, but the pair of cycling transitions has not yet been chosen or optimized.

Compared to hot-wire or absorption techniques, LIF has the advantage of observing a signal against a dark background (modulo light scattered from the optics and vacuum chamber windows). [68] Furthermore, since the emitted light can be collected at different angles, we can obtain spatial information about the detection process. [69]

II. SYSTEMATICS

In general, the statistical sensitivity of a molecular EDM experiment depends on electric polarization P, coherence time τ, detection rate $\dot{N}$, total measurement time T, and other factors encoding the intrinsic sensitivity of the system. For measurements of the Schiff moment in a nucleus with mass A and atomic number Z, the sensitivity scales as $P\tau\sqrt{\dot{N}T}AZS_0$, where S_0 depends on the details of the nuclear structure. [70]

The experiment is expected to result in a null measurement of effects that are odd under T-reversal. A crucial part of the experimental design is to make sure there are no known effects that could mimic genuine T-odd interactions. During data taking, we plan to continually reverse a number of parameters that in principle should not affect the measured NMR frequency f if done perfectly, and measure the resulting shift

$$S_{A,B,\ldots} \equiv f(A, B, \ldots) - f(-A, -B, \ldots).$$

We note that in previous experiments [27], it was also important to consider the shifts in beam intensity corresponding to parameter reversals. In contrast, we are able to measure both spin-up and spin-down populations, making us insensitive to changes in beam intensity and the resulting systematics. By studying the ratio of the difference and the sum of the two populations, the overall beam intensity is in principle cancelled out unless the intensity change is different for the two NMR states, or for the states they connect to in the state preparation regions around the main interaction region. We have not yet found that to be the case.

The following paragraphs discuss the frequency shifts in the order of increasing number of simultaneously reversed fields. See Table II for a list of parameters that we are considering including the sequence of reversals.

A. Reversals of one parameter

The separation between hyperfine states in TlF is a function of the external electric field, but not of its sign. A non-reversing part of the field, always present at some level, can turn that into a systematic error: at 30 kV/cm, the level spacing changes by order 31.5 mHz if the ± 30 kV/cm fields differ in magnitude by 1 V/cm, as may be calculated numerically by diagonalizing the ground-state Hamiltonian in the presence of the two near-opposite fields (see Fig. 9 and Table I).

The shift S_{int} is thus a sensitive measure of how perfectly the interaction-region electric field is reversed. However, a measure independent of the EDM signal is useful. The ACME II experiment characterized the electric field in the interaction region via microwave spectroscopy of a state susceptible to large Stark shifts; see [71, sect 5.1] for details.

In addition to a DC offset on the high-voltage electrode supply, or the finite charging time of the parallel-plate capacitor formed between the electrodes, the incomplete reversal of E_{int}

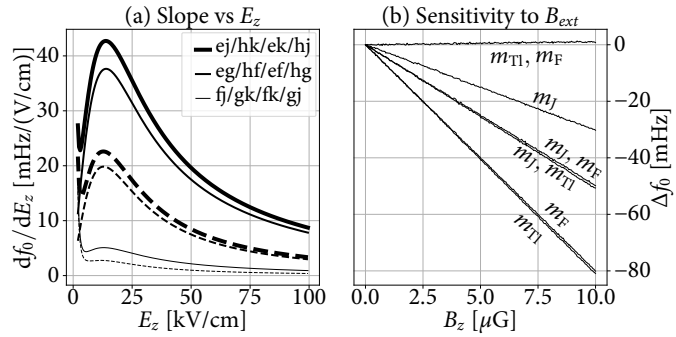

Figure 9: ***a.*** Slope of the NMR resonance frequency f_0 in the applied electric field as a measure of the experimental sensitivity to imperfections in the field reversal. The dashed lines are calculated with the c_3 term from ref. [31], whereas the solid one uses the c_3 from eqn 34. See Table I for the meaning of transition labels. ***b.*** Difference in transition frequency between the T-reversed pairs of transitions, labelled by the spins that get flipped.

can be due to accumulations of charges on non-conductive parts of the electrodes, their mounting structure, or the vacuum chamber itself. The charge accumulations are difficult to predict and will have to be verified experimentally from the energy shifts once the apparatus is built. In ref. [31], they were able to compensate only partially for the charge accumulations by placing a variable DC offset on the electrodes.

The electrode charging time should be estimated once the setup is in place, and can also be extracted from the difference ΔS_{int} in the shifts S_{int} where E_{int} is switched from positive to negative, compared to the negative-to-positive shift. This shift was in fact one of the larger effects in ref. [27], but the authors chose to suppress it with further reversals instead of trying to locate and remove its cause.

It is also possible that the field reversals gradually change the field magnitude over time. That would be evident from plotting S_{int} over time, unless another contrary effect were to conceal the gradual drift in $|E_{\text{int}}|$. Indeed, any other permutations of the switching sequence could cause systematic effects of their own, and will have to be investigated.

The polarity of the field in the quadrupole should not impact the operation of the lens since it works on the principle of a quadratic Stark effect. For reasons analogous to those in the preceding paragraph, however, it will be important to verify this experimentally. In addition, we can expect part of the quadrupole field to leak into other part of the apparatus, in particular the interaction region. A small fraction ε of the leaking reversing quadrupole field $\mathbf{E}_{\text{quad}}$ interacting with a constant precession field $\mathbf{E}_{\text{int}}$ would effectively modify the absolute magnitude of the latter:

$$|\mathbf{E}_{\text{int}} + \varepsilon\mathbf{E}_{\text{quad}}| \neq |\mathbf{E}_{\text{int}} - \varepsilon\mathbf{E}_{\text{quad}}|, \quad \varepsilon \ll 1.$$

In TlF, there is an internal magnetic field at the site of the thallium nucleus due to the rotational motion of the molecule

[27]. By flipping the sign of the projection m_J, this internal field can be reversed exactly. The reversal is not expected to result in a change of the hyperfine level separation unless the external fields change in magnitude, such as due to leaking electric or magnetic fields from the state-selection chamber. However, the residual magnetic field in the interaction region, coming from an imperfect demagnetization of the magnetic shields, would form a sum or difference with the internal molecular fields, such that the absolute magnitude of the magnetic field at the site of the thallium nucleus would change depending on the sign of m_J.

To estimate the magnitude of the internal magnetic field, compare the magnitude of the the Zeeman shift due to the Tl nucleus $-(\mu_1/I_1)(\mathbf{I}_1 \cdot \mathbf{B})$ to the strength of the Tl spin-rotation energy $c_1(\mathbf{I}_1 \cdot \mathbf{J})$. The two effects are equal in strength at $B_{\text{int}} \approx 50.8$ G. For an expected residual external field of $B_{\text{ext}} \approx 10\ \mu$G, we expect S_m of order 80 mHz. In ref. [31], they were able to reduce the S_m greatly adding large Helmholtz coils around the interaction region to cancel out the residual field. Nonetheless, fluctuations of 10-100 mHz persisted.

While we will be careful to develop the procedure for demagnetizing the magnetic shields before they are installed, it can be challenging to verify the residual field once the vacuum chamber with the field plates is placed inside the shield. Instead, we have to use the TlF molecules themselves as magnetometers. For example, a superposition of the ^{19}F spin up and down states would in principle give a pure magnetometer signal in contrast to the analogous ^{205}Tl measurement, sensitive to the EDM signal. A further test—averaged over all directions of E_{int}— of the residual field can be extracted from the ^{205}Tl signal while simultanously reversing the Helmholtz coil currents and/or the internal molecular magnetic field via m_J.

The relative phase of the separated oscillatory fields is switched between $\pm\pi/2$. In either case, if the two thallium spin projections acquired no relative phase delay during the molecular precession time, we expect to read out two equal populations. Indeed, ref. [31] found no significant effects on the EDM shift arising from the nominally $\pm 1°$ phase accuracy between the two coils. However, in CENTREX we are aiming for a more precise measurements, and it is worthwhile estimating the allowed phase error. *TODO: Calculate the resulting signal dependence on beam velocity if the phase shift is not zero.* In addition, we can also switch the cables used to connect the coils to their sources—their different lengths will cause different phase shifts. *TODO: You might read the old Tl atom EDM papers and/or theses to see what phase shift errors were found to do there–it's so long ago now that I don't remember. I have a copy of Chris Regan's thesis (last student on Tl atom EDM) somewhere if you can't find it.*

Table II: Parameters that can be reversed to check for systematics. The symbol in the first column labels the shift corresponding to the reversal of the given parameter; see text for a definition of these shifts. Shifts listed as potentially present depend on as-of-yet unfinished apparatus. The high-voltage reversals (S_{int} and S_{quad}) can be done either automatically via the HV supplies, or manually by switching the leads.

Symbol	Param.	Description	Values
S_{int}	E_{int}	interaction-region electric field	± 30 kV/cm
S_{quad}	V_{quad}	quadrupole lens voltage	± 30 kV
S_m	m_J	projection of molecular rotation	± 1
$S_{P1,2}$	$P_{1,2}$	relative phase of coils 1 and 2	$\pm\pi/2$
S_{F}	M_{F}	spin-projection of fluorine nucleus	$\pm 1/2$
Potentially present:			
S_{Bss}	B_{ss}	state-selection magnetic field	?
S_{Ess}	E_{ss}	state-selection electric field	?
S_{p1}	$\lvert p1\rangle$	initially populated NMR state	?
S_{d}	$\lvert d\rangle$	detection final state	?

B. Two-parameter couplings

If the state-selection region will feature a static magnetic field B_{ss}, used to select one of the $m_J = \pm 1$ states, the shift S_{ss} due to the interaction of the leaking state-selection field and the internal molecular will vanish when averaged over the m_J modulation since m_J determines the direction of the internal magnetic field. Consequently, it will be important to measure the shift $S_{\text{ss},m}$ when both m_J and B_{ss} are reversed simultaneously.

A further shift under simultaneous reversal of B_{ss} and m_J is due to the so-called Millman effect [72], due to the spatial curvature of the fields inducing the NMR transition. The oscillating field can be decomposed into two rotating fields, and only one of them is actually inducing the transition, depending on the direction of the magnetic field. In the frame of the molecule, the frequency of the oscillating field changes with the curvature of the fields, resulting in B-dependent change in the resonant frequency. [31] For two separated coils, to first order the shift due to the spatial curvature of the field is proportional to the geometrical angle between the coils, and inversely proportional to the time of flight T of the molecules between the two coils, resulting in a frequency shift $\theta/2\pi T$. [27, eqn 12]

The Millman effect could be distinguised from an ambient magnetic field by a beam velocity reversal (impractical in the CENTREX setup), or by varying the beam velocity so as to change T. In a pulsed beam, the natural velocity spread means that the time of detection is strongly correlated with the velocity. By analyzing data separately for different velocity classes, we should be able to find the extent of the Millman effect. *TODO: Quantify this based on data we have about velocity distribution and temporal width of beam pulse, but I think the correlation will be very strong because our beam is so long.*

It is possible that the reversal of E_{int} gives rise to a magnetic

field, either due to the electronics effecting the reversal, or due to currents leaking between the field plates. In ref. [27], they employed an additional field reversal by manually rewiring the electrode high voltage supply (motivated by the switching relay's frequent breakdowns [39]), finding that the switching electronics does not noticably impact the EDM measurement (at least, the electronics are no worse than the manual electrode polarity switching). To guard against the second type of induced magnetic field, they made sure that the leakage current remained below 2 nA at the operating voltage, an entirely negligible effect. In CENTREX, it will also be important to verify that the HV electronics, and the plate charging currents, do not incur significant systematics. *TODO: Estimate how large a leakage/charging current is tolerable for us, and what would be needed (in terms of charging time and/or effective resistance) to keep things below that limit.*

A further magnetic field correlated with the electric field comes from a Lorentz transformation into the frame of molecules moving with a velocity $\mathbf{v}$ through a large electric field. The resulting motional magnetic field has the value $\mathbf{B}_{\text{mot}} \approx \mathbf{E}_{\text{int}} \times \mathbf{v}/c^2$. However, since the internal molecular magnetic field is exactly parallel to the applied electric field, $\mathbf{B}_{\text{mot}}$ does not give rise to a first-order energy shift. The second-order shift, being quadratic in E_{int}, is not a dangerous systematic unless there is a significant residual magnetic field present in the interaction region. That field can sum with the motional field to a total field not exactly perpendicular to E_{int}, giving cross terms between the two fields that change sign with the applied electric field. *TODO: Give quantitative calculations of effect size vs. imperfection sizes: the issue is what is the total B-field and its direction relative to E_{int}. Then, use that to figure out how the energy difference between states of interest changes when you reverse E_{lab}.*

Any currents in the vicinity of the magnetic shield can also magnetize the magnetic shield itself, coupling together seemingly unrelated parameters. For example, currents for charging the high-voltage electrodes in the interaction region or the lens could do this, and cause a residual magnetic field that reverses with the applied electric fields. *TODO: It would be good to estimate crudely how large this effect could be, and how much of a problem it would be at that level. If the answer is "a lot", then thinking of measures to reduce it will be important. The 199Hg EDM experiment (papers, theses) will likely give some useful information about what has been achieved, since the parameters (eg voltage, capacitance) are somewhat similar to ours.* For that reason, the shields will have to be degaussed regularly.

TODO:

- 'Bloch-Siegert' shifts (AC Zeeman shifts from imperfections in the RF drive fields, in combination with various E-field imperfections): Bloch-Siegert is specifically the AC Zeeman shift caused by the counter-rotating term you ignore when you do the rotating wave approximation. So, it induces shifts proportional to the RF drive power. You can determine their effect by changing the RF power–for example, drive a little above vs a little below the power needed for a pi/2 pulse. The Ramsey signal changes only quadratically with the deviation from ideal power [I think...check!] while the BS shift [heh] is linear in power. In an extreme case, we could in principle drive a 3pi/2 pulse instead of pi/2 to check on its effect.

- Re the analogous measurement with 19F (no new physics, all same systematics): First, see if you can find a scheme to populate and detect a pair of states like this. Make a list of all the control parameters that would have to change (RF frequency of course, also microwave frequencies in state selectors...anything else?) The main point is to look for magnetic field effects, but generally speaking any estimate of systematics with 205Tl should be revisited here, to see which (if any) contribute at about the same level vs. quite different levels, e.g. E_{dc} induced freq shift might be quite different here.

- It would be a good idea to think about geometric phase effects also. Useful information in Tl atom theses/publications, also ACME (Amar Vutha thesis, maybe also Brendon O'Leary's?). I don't think this has been analyzed for TlF at all.

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- Maybe some effect of oscillating RF E-field due to Maxwell $[curlE \propto dB/dt]$? Can it drive transitions or induce energy shifts like B-S shifts? Not sure about this one, but scratching out crude numbers would be useful.

- For each effect that I have estimated in terms of the frequency shift it causes, calculate how much we need to suppress it to get an acceptable final accuracy. Compare this to how well has been done in other experiments. If we need to do something many orders of magnitude better than others have done after trying hard, we are probably in trouble and need a better plan. Look through other literature (Tl atom EDM, 199Hg EDM, neutron EDM) to get a sense of what has been accomplished r.e. leakage currents, E-field reversibility, stray magnetic fields, etc.,

- read some ACME theses (Brendon, Zack, Cris Panda) to get a sense of how we look for systematics there

- Continue to think about ways we can pull relevant information out of data with the molecules.

- Think about what controls we will need that we may not have been thinking about. Obviously, ability to apply small B_z field in interaction region. Probably also small B_y and B_x. Almost certainly also small gradients e.g. dB_z/dx. How many precision current supplies will we need? What about other things like RF generators–what is the best we can find r.e. phase shift accuracy? etc.

III. SIMULATIONS

A. Rotational cooling

TODO: TBD.

B. Non-adiabatic transitions

Molecules travelling from the first state preparation region into the electrostatic lens experience fast changes in external electric field. Due to the large number of hyperfine levels involved, predicting level stability is a non-trivial task. As a consequence of the von Neumann–Wigner rule [73], levels of same symmetry will in general avoid crossing each other. Considering only a single such avoided crossing between a pair of states, we may model the situation using the two-level Hamiltonian

$$H = \begin{pmatrix} \delta - \alpha\varepsilon^2 & \beta \\ \beta & 0 \end{pmatrix}, \tag{49}$$

where $\delta \propto 2\pi \times 100$ kHz is the field-free level separation, $\alpha\varepsilon^2$ is the level shift due to a quadratic Stark effect; thus α sets the field ε at which the Stark effect cancels δ (typically, tens or hundreds of V/cm). Finally, β is the level coupling that gives rise to the avoided crossing.

The Hamiltonian H may be diagonalized exactly to give the eigenenergies $E_\pm$ and states $\psi_\pm$. To investigate the effect of subjecting the molecule to a time-varying electric field $\varepsilon(t)$, we can assume that at $t = 0$ the system starts in the state ψ_+. After solving the time-dependent Schrödinger equation (TDSE) $i d\psi/dt = H\psi$ for the final state ψ_f, we can evaluate the probability $P_\mathrm{trans} = |\langle\psi_\mathrm{f}|\psi_-\rangle|^2$ the system has transitioned to the other field-free eigenstate.

If the electric field is chosen such that the system is taken through the avoided crossing in a time T_fall, then the transition probability $P_\mathrm{trans}(T_\mathrm{fall})$ approaches zero in both limits $T_\mathrm{fall} \to 0$ and $T_\mathrm{fall} \to \infty$ (see Fig. 10a). For intermediate times, the system exhibits a pattern of interference fringes, with P_trans very sensitive to the exact timing of the electric field changes.

The idea of the preceding paragraph finds application in the problem of molecular motion through regions of spatially-varying electric of magnetic fields. For instance, in CENTREX we prepare molecules in the particular hyperfine state that is expected to be focused best by the hyperfine lens. However, the molecular beam has a range of velocities, such that different molecules will experience a different rate of change of the external fields, with some non-adiabatically transferred to a state other than the lens state, preventing and defeating the focusing otherwise expected of the lens.

Adding to the above a rapidly-oscillating AC electric field can under certain conditions suppress the transitions otherwise expected in the absence of the AC field. A plot of P_trans with respect to the parameters of the AC field (Fig. 10b) shows separated regions of high P_trans against a transition-free background. Furthermore, one of the regions does not show dependence on the frequency ω of the oscillating field, while the other regions show what appears to be polynomial dependence on ω.

The above results can be qualitatively be interpreted in terms of Floquet theory for 'stationary' states of periodically driven systems [74, ch. 5]. It can be proven that the TDSE of such a system has solutions of form [75, sect. II.A]

$$\exp(-i\varepsilon_\alpha t\hbar)\Phi_\alpha(x,t),$$

introducing the periodic Floquet modes $\Phi_\alpha(x,t)$ which satisfy a 'time-independent' Schrödinger equation (TISE) in the extended Hilbert space $L_2[0,T] \otimes H$

$$\underbrace{\left[H(t) - i\hbar\frac{\partial}{\partial t}\right]}_{\widetilde{H}} \Phi_\alpha(t) = \varepsilon_\alpha \Phi_\alpha(t),$$

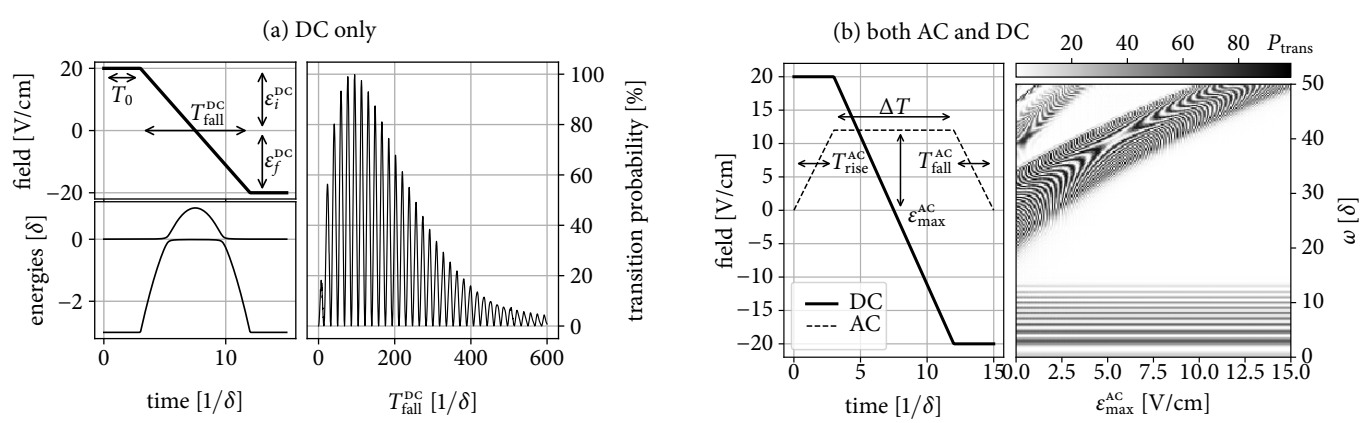

Figure 10: AC level stabilization against non-adiabatic transitions when subjected to time-varying electric fields. $\varepsilon_\mathrm{i}^\mathrm{DC} = -\varepsilon_\mathrm{f}^\mathrm{DC} = 20$, $\alpha = 0.01$, $\beta = 0.1$, $\delta = 1$. ***a.*** The transition-probability interference fringes as a function of field reversal time, with the time dependence of the field and the eigenenergies illustrated in the left plots. $T_0 = 8$, $\Delta T = 2$. ***b.*** Adding an AC field results (magnitude shown in left plot, dashed line) in separated regions of high P_trans against a transition-free background (right plot). $T_0 = 1$, $T_\mathrm{fall}^\mathrm{DC} = 196.6$, $\Delta T = 200$, $T_\mathrm{rise}^\mathrm{AC} = T_\mathrm{fall}^\mathrm{AC} = 1$.

To obtain the matrix elements of the Hamiltonian thus modified, we expand the Floquet modes in a basis $\{|k\rangle\}$ of H, and also in terms of nth order plane waves $e^{in\omega t}$:

$$\Phi_\alpha(t) = \sum_k \sum_{n=-\infty}^{\infty} c_\alpha^{kn} e^{in\omega t} |k\rangle.$$

Insert into the TISE, pre-multiply by $\langle j|e^{-im\omega t}$, and time-average over one cycle to obtain

$$\sum_{kn} \langle j|\widetilde{H}|k\rangle c_\alpha^{kn} = \varepsilon_\alpha c_\alpha^{jm},$$

where the matrix elements of the extended Hamiltonian $\widetilde{H}$ are given in terms of the original Hamiltonian split into constant and time-varying parts $H(t) \equiv H_{\mathrm{DC}} + H_{\mathrm{AC}}(t)$:

$$\begin{aligned} \widetilde{H}_{nm}^{jk} &= H_{\mathrm{DC}}^{jk}\delta_{nm} + n\hbar\omega\delta_{nm}\delta^{jk} && \text{(atom + light)} \\ &+ \frac{1}{T}\int_0^T H_{\mathrm{AC}}^{jk}(t) e^{i(n-m)\omega t} dt. && \text{(interaction)} \end{aligned}$$

Here, the lower indices refer to a particular 'photonic' mode, and the upper indices refer to the matrix elements of the original Hamiltonian. This way of separating the Hamiltonian into the atomic, radiation, and interaction operators resembles the 'dressed state' picture for a two-level atom in a monochromatic field of radiation (cf. eqn 1.18 in ref. [76]); the above Hamiltonian appears to provide a generalization beyond the two-state model, since $H_{\mathrm{AC}}^{jk}(t)$ can be of any dimensionality.

The interaction term in the above general expression can easily be found for the case of the two-level model in eqn 49. When an electric field $\varepsilon(t) \equiv \varepsilon_{\mathrm{DC}} + \varepsilon_{\mathrm{AC}}(t)$ is applied, the AC part of the Hamiltonian is

$$H_{\mathrm{AC}} = -\begin{pmatrix} \alpha & 0 \\ 0 & 0 \end{pmatrix} \left(2\varepsilon_{\mathrm{AC}}\varepsilon_{\mathrm{DC}} \cos\omega t + \varepsilon_{\mathrm{AC}}^2 \cos^2 \omega t, \right)$$

which may be time averaged to find the AC matrix elements

$$\begin{aligned} -(H_{\mathrm{AC}})_{nm}^{00} = \Bigg[&\varepsilon_{\mathrm{AC}}\varepsilon_{\mathrm{DC}}(\delta_{n,m-1} + \delta_{n,m+1}) \\ &+ \varepsilon_{\mathrm{AC}}^2 \left(\frac{\delta_{n,m}}{2} + \frac{\delta_{n,m+2}}{4} + \frac{\delta_{n,m-2}}{4} \right) \Bigg]. \end{aligned}$$

The Kronecker deltas give the order of the atom–photon interaction. For instance, the 0th order term proportional to $\delta_{n,m}$ modifies the Hamiltonian to the effective form

$$H_{\mathrm{eff}} = \begin{pmatrix} \delta - \alpha\left(\varepsilon_{\mathrm{DC}}^2 + \varepsilon_{\mathrm{AC}}^2/2\right) & \beta \\ \beta & 0 \end{pmatrix};$$

i.e., the 0th order effect of the AC field is a simple AC Stark shift, where the RMS AC field has the same effect as a DC field. This suggests that the mechanism through which the AC field suppresses transitions is by limiting the effective miniumum value of the DC field, such that the system no longer passes through the avoided crossing.

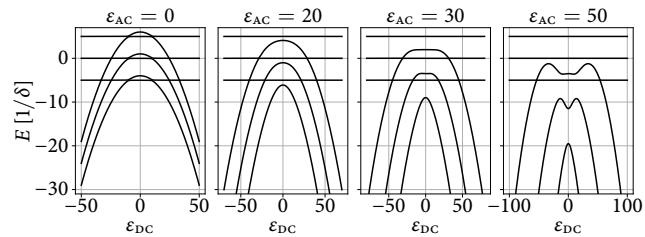

Figure 11: Eigenstates of the 6×6 Hamiltonian $\widetilde{H}_{nm}^{jk}$, where n, m are the indices of the original 2×2 atomic Hamiltonian, and j, k here limited to the $\{-1, 0, 1\}$ part of the infinite Hilbert space. Without the AC field, note that the spectrum resembles that of the atomic Hamiltonian (as in Fig. 10), thrice replicated. As the AC is turned stronger, the location of the avoided crossings moves around or disappears.

The higher order interactions—due to the $\delta_{n,m\pm1}$ and $\delta_{n,\pm2}$ terms—expand the Hilbert space of the two-state Hamiltonian, effecting a tensor product with the corresponding Kronecker matrices. The effect on the spectrum is to replace the two states with an infinite set of pairs of states, each pair separated from the next by the energy $\hbar\omega$ of a single photon (Fig 11). Plotted as a function of the electric field, we note that in addition to the avoided crossings present in the 'unperturbed' Hamiltonian, new crossings appear as a result of interactions with the virtual photons.

The new crossings typically appear at higher fields than the original ones and thus do not cause transitions unless the inverting DC field is of sufficient magnitude to reach them. In the simulation resulting in Fig. 10b, the DC field is assumed to traverse the inherent avoided crossing and not much further. However, in the flight of molecules towards the electrostatic lens, fields of order 30 kV/cm get reversed in a characteristic time $T_{\mathrm{fall}} \propto 10^{-4}$ s $\propto 10/\delta$. In order to move the induced avoided crossings beyond the reach of these fields, the applied AC field would need to have a frequency of order 900 GHz. Alternatively, the AC field could be confined to a region where the DC field does not depart too far from zero.

C. Optical cycling

TODO: TBD.

IV. APPARATUS

As detailed in the section on the measurement scheme, CENTREX follows the design of a conventional molecular beam experiment. Fig. 12 displays an outline of the entire apparatus, Table III contains key parameter values, and the following paragraphs give information about its construction.

A. Control software

The experimental parameters are monitored and controlled using a range of devices. Most of these have already been integrated into a central data acquisition (DAQ) system. At present, the latter consists of a single highly configurable Python program running on all computers attached to the experiment, communicating with the devices connected to the computer and writing the obtained data to the local disk. In the future, we intend to set up a common data and backup server to deal with the greatly increased data generation rate once the entire apparatus is up and running.

The program's code is divided between classes that make up the graphical user interface (GUI), and control classes which make use of driver classes to communicate with physical instruments. In addition, all program configuration is contained in `.ini` files in subdirectories of the so-called `config` directory, allowing for simple modification and extension of the DAQ system without changing the main program code. This configuration is loaded into the relevant `Config` class.

The user controls the devices through the `ControlGUI`, which is a canvas for control of recording and external devices. The information in device config files is automatically read and translated into usable controls that appear in the GUI. Underneath the graphical interface, several classes implement the program's functionality:

(1) `Device` objects run each within its own thread, instantiate device drivers, and poll devices at regular intervals for data as well as check for normal operation insofar as supported by the driver, pushing this information into data and events queues specific to each device. In addition, these objects serve as an unified abstract interface to all the drivers and related information, especially through the `config` member (of type `DeviceConfig`).

(2) `Monitoring`: While the `Device` objects collect information from the physical devices, they neither record it nor notify the user in case of any abnormal condition. Thus the main program instantiates one `Monitoring` object to read some of the data, display it in the main interface, and send it to an external central database.

(3) The `HDF_writer` stores the data collected by `Device` objects to an file for future reference and analysis. The files are of Hierarchical Data Format (HDF), version 5 format [77], a portable, cross-platform format that provides APIs for commonly scientific computing languages. The format is extensible, open, and structured, allowing storage of data and its description in terms of arbitrary attributes, column names, and groups.

(4) `Config` classes serve to organize access to program, device, or plot configuration. Thus, instead of having classes pass ad hoc pieces of information between each other, they set an appropriately-named attribute of the relevant `Config` class. The latter in term only allows to set the attributes that have been declared previously. This enables the programmer to have a clear overview of what configuration parameters exist by simply inspecting the relevant `Config` class.

The program is thus generic enough to make it easy to add

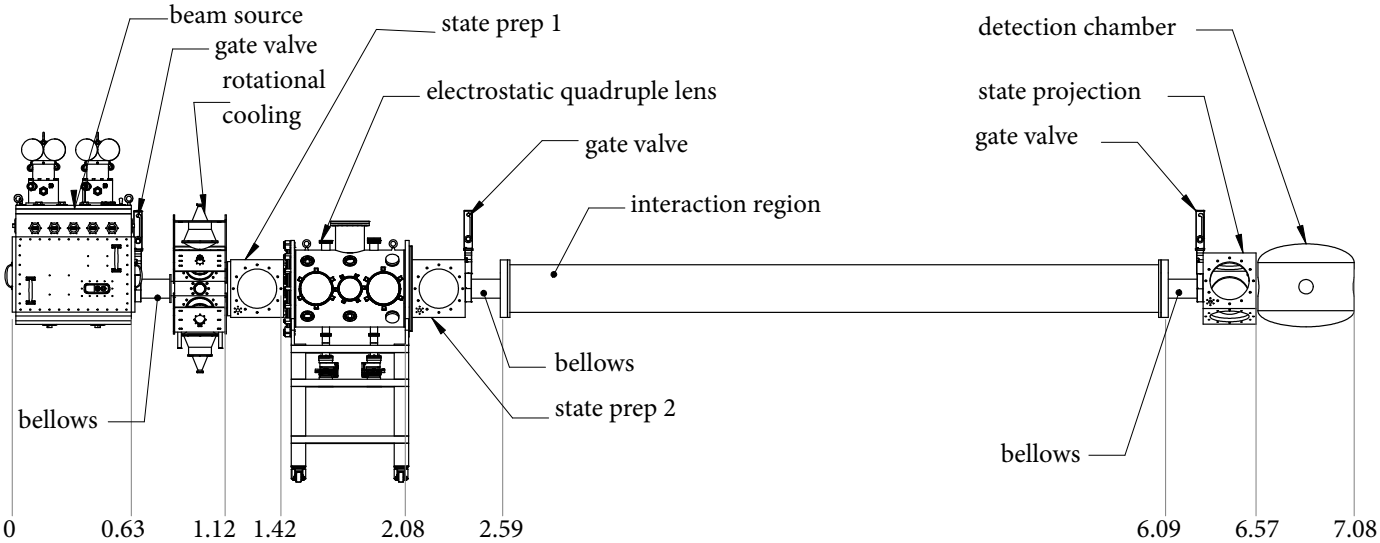

Figure 12: Overview of the CENTREX beamline without the magnetic shield around the interaction region. Note that the second state-preparation region, and the detection chamber, have not yet been designed. The horizontal ordinate is given in meters.

Table III: Key parameters of the CENTREX experiment.

Parameter	Value	Units
Beam source:		
total source intensity	10^{10} to 10^{12}	mol./s/sr/state
flux after rot. cooling	???	mol./sec
average beam velocity	184 ± 17	m/s
beam velocity spread	16.1 ± 0.8	m/s
rotational temperature	6.5 ± 0.2	K
vibrational temperature	8	K
beam diameter	2.5	cm
nozzle diameter	1/4	in
nozzle to detector dist.	6.45	m
Quadrupole lens:		
quadrupole voltage	± 30	kV
bore diameter	1.75	in
length	60	cm
Interaction region:		
SOF coil frequency		kHz
slope of SOF resonance	0.094	1/Hz
interaction region E field	± 30	kV
interaction region B field	< 10	μG
SOF coil separation	3	m
interaction time	16	ms
resonance FWHM	33.2	Hz
SOF coil width	10	cm
Detection region:		
detection efficiency	100 %	—
source-detector distance	594 ± 10	cm
background count	???	mol./sec
detection area	???	mm^2
signal to noise ratio	???	—
fringe contrast	???	—

capabilities to control any number of new devices by simply (1) writing a device driver, and (2) a device config file. Drivers are classes inside the Python modules, such that instantiating the driver class is the abstract representation of opening a connection to a physical device. In addition, a driver class provides a `ReadValue()` method that implements the particulars about how reading data works, and should return a list of values, consistent with the `dtype` and `shape` of the device as specified in the constructor. The driver must also provide a function `GetWarnings()` that is be called at regular intervals to populate the list of `warnings`. This function can check for the device parameters are within normal ranges, or simply do nothing if appropriate.

Upon starting, the program reads the main config file that defines general program settings; the values are read by the instance of `ProgramConfig` class. Device configurations are read from `.ini` files in the chosen directory. Thus choosing a different directory allows for a different set of devices or device configurations to be loaded. These files define the controls that to appear on the screen for each instance of a device, the driver commands called from these controls, and their default values.

The device `.ini` file should specify whether the device is a slow or a fast device. A slow device is expected to return only a few values each time its `ReadValue()` is called, and thus requires a single dataset to contain all the data. A fast device can return thousands or millions of datapoints each time it is polled, together with metadata that describes this particular set of values. Thus, the program creates a new dataset for each invocation of `ReadValue()` of a fast device.

B. Beam source

The beam source works by ablation of a solid sample of TlF held at cryogenic temperatures in a thermally shielded vacuum chamber. The ablated particles are expelled from the ablation cell by a flow of cold neon buffer gas. The design has been adapted from the beam source used in ref. [41] with significant modifications.

The ablation cell is a 1.500 in high cuboid with top and bottom square faces with a 2.000 in side, machined out of OFHC copper (Cu 101). The internal volume consists of two intersecting cylinders of diameter $d = 1.150$ in and length $L = 2.000$ in. According to a formula known to Archimedes [78], the volume common to the two cylinders is $\frac{2}{3}d^3$, making the internal cell volume for two cylinders to be $V = 2L\pi\frac{d^2}{4} - \frac{2}{3}d^3 \approx 3.14\ \text{in}^3$. The actual internal volume is slightly larger than that due to the way windows are attached to the cell.

Three of the four circular openings in the cell are closed with fused silica windows, while the forth is covered by a thin plate of OFHC copper (0.050 in thickness) with a 1/4 in aperture as the molecular beam exit nozzle. Unlike similar beam sources, the exit aperture is *not* followed by a conical collimator, since we found it interferes with the collision-mediated beam thermalization processes (an effect highly sensitive to the exact placement, shape, and dimensions of the collimator). In order to protect the window through which the ablation laser (Fig. 13) enters from the immediate ablation by-products, the window is offset by a 3 in-long 'snorkel', a cylinder of outer diameter 1.250 in, and an hole with internal diam. of 0.9 in going through the entire length of the piece. The cylinder is terminated on one end in a way that allows attaching it to the cell, and on another by a fitting for the window.

The cell opening opposite the ablation laser is covered by the target holder, a copper piece machined so as to contain a volume of solid TlF. The targets are prepared in a separate chamber by filling them with powdered 99 % pure thallium(I) fluoride (supplied by Strem Chemicals, Newburyport, MA, under product number 93-8106) and heating to a temperature

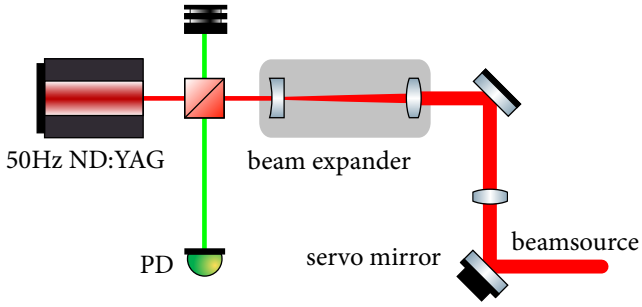

Figure 13: The ablation laser for the beam source is a pulsed Nd:YAG, provinding 4 ns pulses of maximum energy 130 mJ at max. 50 Hz (Nano LG 130-50 by Litron Lasers Ltd, UK). In beam source tests, we have commonly run it such that the pulse energy is approx. 35 mJ, at a repetition rate of 5–10 Hz. The photodiode receives a small amount of light split off the main beam, and is used for monitoring the laser power. [Drawing by O. Grasdijk]

just above the melting point 327 °C. In the early stages of beam characterization, the target holder had a small (1/4 in diam.) opening, covered by another snorkel and window, to allow for absorption probing the TlF population in the cell.

The buffer gas is 99.999 % pure neon, supplied by TechAir (Naugatuck, CT). Outside the vacuum chamber, neon flow is moderated by a 412 series dual-stage pressure regulator (Concoa, Virginia Beach, VA), and a mass-flow controller (MKS Instruments, Andover, MA; part number 1179C00212CR1AV) that allows up to 100 sccm of flow. The gas is fed from the bottle using 1/4 in copper tubing.

The opening opposite the cell exit nozzle has a 0.375 in opening that allow optical access along the direction of the molecular beam. In addition, perpendicularly through the same hole, the neon buffer gas enters the cell through a 1/16 in opening. The gas enters the beam source vacuum chamber via a KF40-to-VCR (1/8 in) feedthrough (Accu-Glass Products, Valencia, CA; part number 113030). To thermally separate the room-temperature vacuum chamber from the cryogenic parts, the connection is insulated by smooth-bore seamless 304 stainless steel tubing, 1/8 in OD, 0.02 in wall thickness (McMaster-Carr Supply Company, Elmhurst, IL; part number 89895K711). The stainless tubing is approx. 3 ft long, wound in a coil (approx. diam. 3 in) to save space. The gas line is thermally connected to the 40 K shield by flowing through 1/8 in copper tubing wound around and brazed on a 3/4 in-diam. copper bobbin that is fixed to the 40 K shield with a threaded rod. Another similar coil of stainless tubing limits heat flow from the 40 K shield to the cell. Finally, the neon thermalizes to the cell temperature by flowing through an approx. 6 in-long segment of 1/8 in copper tubing that is brazed to the piece that attaches to the cell.

The cell is screwed onto the cold stage of the PT815 cryocooler (made by Cryomech Inc., Syracuse, NY). The cell is shielded from ambient thermal radiation by a two-state heat shield. The outer shield, made of aluminum 1100, hangs from the top plate of the vacuum chamber on stainless threaded rods and consists of a frame with removable covering plates and is thermally connected to the 'warm' stages (approx. 40 K) of both pulse-tube cryocoolers. The inner shield, hanging from the outer one on four 1/4 in stainless threaded rods, is made of 1/4 in thick Cu101 plates, and is connected to the cold stage of the colder PT415 cooler, which cools it to approx. 4–5 K. The 4 K shield has bare openings for optical and molecular access, while the optical ports in the 40 K shield are blocked with loosely attached windows to limit the buffer gas escaping into to rest of the apparatus. The cell–cryocooler thermal links are done via flexible copper braids to prevent damage from unmatched thermal expansion coefficients.

For control of cell temperature, a number of 10–25 Ω resistors are attached directly to the cell and heated directly with standard laboratory power supply, or controlled in a PID loop using the CTC100 cryogenic temperature controller (Stanford Research Systems, Sunnyvale, CA). The same controller gets temperature input from the silicon diode sensors (DT-670 from Lake Shore Cryotronics, Inc., Westerville, OH) located on the cell gas-input nozzle, cell top plate (target side), 4 K and 16 K pulse tube warm stages. The other eight sensors, recorded by the model 218 temperature monitor from Lake Shore Cryotronics, are located on the cell back snorkel, 4 K shield top plate, 40 K shield top plate, 4 K pulse tube cold head, cell top plate, 4 K shield bottom plate, 40 K shield bottom plate, and the 16 K pulse tube cold head. These eight sensors' wires are fed through the vacuum chamber using two 25-pin feedthroughs (Accu-Glass p.n. 100225), with corresponding vacuum-side PEEK female D-sub connectors (Accu-Glass p.n. 100460) for ease of disconnecting.

Cell windows are standard fused-silica UV-grade windows (from Thorlabs, Newton, NJ). The anti-reflect coatings, done at 271 nm (excited-to-ground molecular transition) and 1064 nm (ablation laser) were done by Thin Film Lab (Milford, PA). All removable mechanical connections are done by means of brass screws into threaded holes, reinforced with stainless threaded inserts (Heli-Coil). All gas-line connections are done with VCR fittings brazed onto the copper or stainless tubing. Where possible and desirable, metal-to-metal thermal conductivity has been enhanced either with indium foil, or vacuum-compatible grease (Apiezon N).

C. Pumping system

In addition to providing thermal isolation, the shields also serve as cryopumps, lowering the pressure from $\sim 10^{-6}$ torr, as can be reached by turbomolecular pumps, to approx. $2 \cdot 10^{-8}$ torr. Pumping down from atmospheric pressure is done with nXDS 6i and 10i dry scroll pumps (Edwards Vacuum, UK) and HiPace 700M turbomolecular pumps (Pfeiffer Vacuum Technology AG, Germany). Pressure is monitored with ionization vacuum gau-

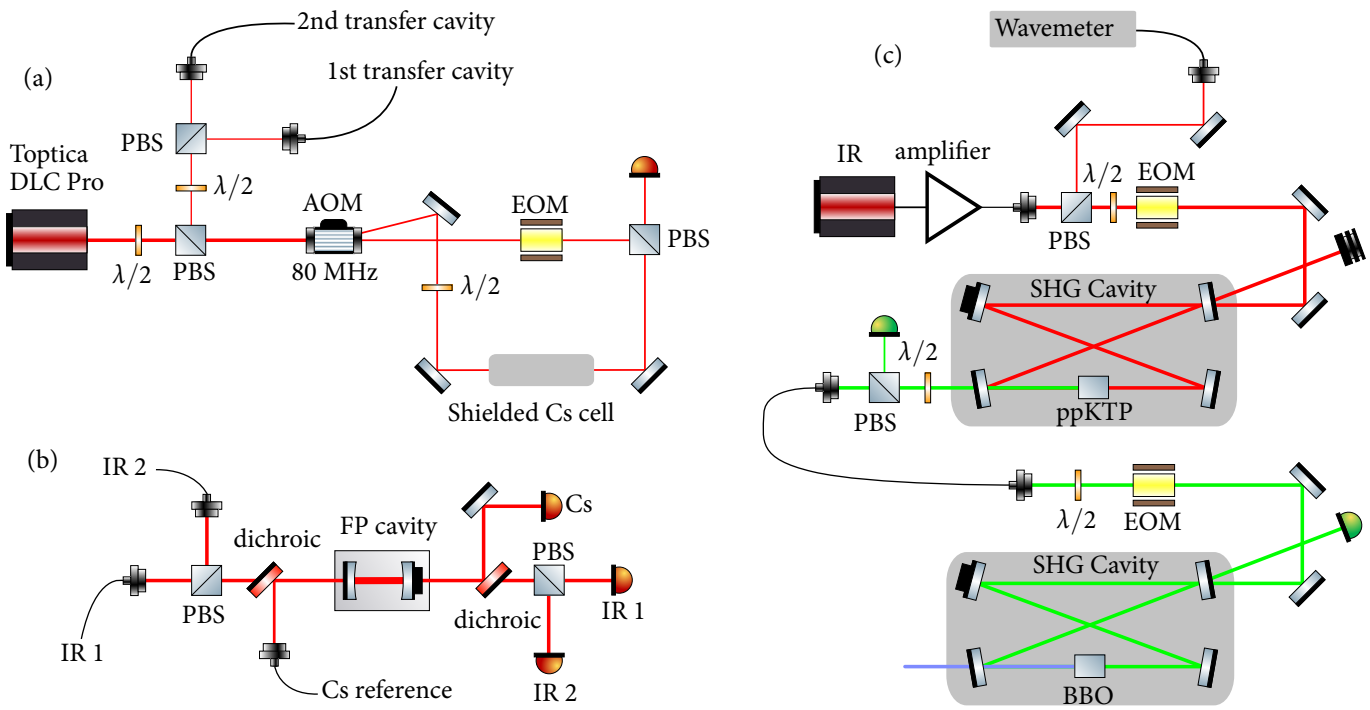

Figure 14: UV laser system (see text). ***a.*** Cs reference laser. ***b.*** Transfer scanning cavity frequency lock. ***c.*** Two-state second harmonic generation (SHG). [Drawings by O. Grasdijk]

ges (IGM401 Hornet from InstruTech LLC, Longmont, CO).

D. Laser system

See Fig. 14 for a diagram of the UV laser system. It consists of three systems working together to (1) provide a stable cesium reference line, (2) lock an IR seed laser to the reference, and (3) to quadruple the IR into a UV beam that can be used to probe transitions in TlF, occuring around 271 nm. In order to provide separately-controllable laser frequencies, we have three SHG systems connected to two transfer lock cavities (one of them currently under construction).

Cs lock The reference system is based on modulation transfer spectroscopy (MTS), a pump–probe technique resulting in sub-Doppler resolution for laser locking. An EOM modulates the pump beam (so as to cause sidebands to accompany the carrier frequency) before it gets passed through a vapour cell. Modulation can appear on the counterpropagating probe beam, passed through the same vapour, through the interactions of the two beams with the atomic vapour. The interactions are an example of four-wave mixing where two pump beam frequency components combine with the probe beam, resulting in a sideband for the probe beam. [79]

In contrast, in frequency modulation spectroscopy (FMS) only a single modulated beam is passed through the sample and onto a detector. If the modulation frequency is large compared with the spectral feature being probed, then the laser (or EOM) can be tuned such that only one sideband interacts with the sample. In the absence of such an interaction, the RF signal arising from the beating of one sideband with the carrier cancels that arising from the other sideband. When this balance is upset by phase or amplitude changes in one sideband, and a background-free signal results. [80]

Following ref. [81], the beam sourced from the diode laser (DL pro 850, Toptica, Germany) is split off for transfer cavities with two polarizing beam splitters (with the two half-wave plates controlling the amount of light sent to each transfer cavity beam). The remaining beam is split by the acousto-optic modulator (AOM) (AOMO 3080-122 830NM from Gooch & Housego, UK) with 80 MHz center frequency, AR-coated for operation at 780–850 nm, into a pump and probe beam, the pump having about twice the power of the probe. (The AOM also eliminates trouble from backreflection from the Cs cell, or photodiodes.) The pump beam is phase-modulated by the custom-built EOM (Thorlabs Inc.) at 3 MHz before being led through the cesium quartz cell (Thorlabs GC19075-CS, diam. 19 mm, length 75 mm), magnetically shielded to the milligauss level in a 6 in-long, 2 in inner diam. three-layer concentric mu-metal shield (modified ZG-203 from Magnetic Shield Corporation, Bensenville, IL).

The MTS signal is detected by the photodiode (we have not found it necessary to make use of the FMS), and mixed with the EOM driving frequency to obtain an error signal used inside the laser controller (Toptica DLC pro) to stabilize the laser diode current and the piezo-controlled cavity, and lock it to the $F = 4 \rightarrow 5$ transition in the hyperfine structure of the cesium D_2 line, occuring at 852.35641 nm [82]. However, as the Cs

reference is only used to set the scale in the FSR of the transfer cavity, the choice of the hyperfine level is not important—GHz differences cause differences negligible compared to the ≈ 0.5 MHz stability required for the IR science lasers.

Transfer cavity The Fabry–Pérot interferometer (Toptica FPI 100) with free spectral range (FSR) 1 GHz is locked to the cesium reference (master) laser. Up to two slave lasers can be used in this scheme, one for each of the two orthogonal polarizations, while the master can be distinguished by its different wavelength and separated away with a dichroic mirror. The cavity length is modulated by slighly more than one FSR to scan over the transmission peaks. The timing difference between the master and slave peaks is then fed to a PID loop to lock the slave frequency.

Second harmonic generation The IR beam at 1087 nm is generated by the Koheras Basik fiber laser seed and amplified to $\approx$ 2 W by the Koheras Boostik Fiber Laser (both devices by NKT Photonics A/S, Denmark). A small amount of power is split off for the wavemeter (671A by Bristol Instruments, Rochester, NY), which allows for a measurement of the absolute laser frequency to within 50 MHz. The rest of the beam is modulated by an EOM and passed through a bowtie-configuration cavity (built by K. Wenz at Columbia, NY) containing a ppKTP crystal that generates up to 1 W of green light at 543.5 nm. The green beam is coupled into a fiber and delivered to a second SHG system (Toptica SHG pro) with a similar configuration. A BBO crystal is used for frequency doubling, resulting in $\approx$ 70 nm of UV (271.7 nm).

To counteract the cavity length changes due to temperature and pressure variations, one of the cavity mirrors is attached to a piezo actuator. Using the Pound–Drever-Hall technique (PDH), the cavity photodiode signal is mixed with the EOM driving frequency in order to extract a sensitive error signal that can be used to PID-control the piezo actuator. [83] The UV lock is done with a commercial analog setup (Toptica PDH 110, PID 100, and FALC 110), while for the green cavities we are transitioning to a home-built system, based on the Red Pitaya STEMlab FPGA board.

E. Rotational cooling

We designed a custom aluminum vacuum chamber to allow access for the molecular beam, UV laser, and microwaves as required by the rotational cooling scheme.

The microwave system (see Fig. 15) is designed to provide up to three microwave beams (at frequencies 13.4, 26.8, and 40.1 GHz, and power of order 0.1–1 W), each focused with a quasioptical lens built into the transmitting antenna so as to position the beam waist in the center of the vacuum chamber. In addition, the beams can be switched between either of the two orthogonal linear polarizations at 1–2 MHz. In order to be

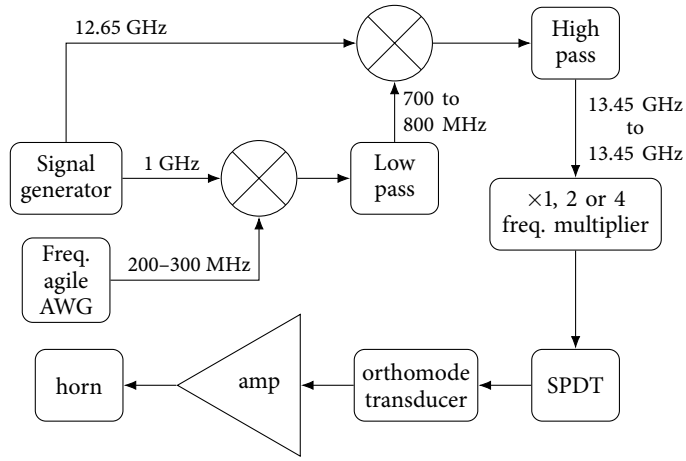

Figure 15: The principle behing frequency-agile rotational cooling microwave generation. The frequencies are shown assuming the ×1, 2 multipliers for a horn output of 13.4 and 26.8 GHz. For 40.1 GHz, the system is adjusted to generate approx. 10 GHz (pre-multiplier).

able to separately address the hyperfine levels in the molecular ground state, the frequency can be rapidly adjusted by means of direct digital synthesis (DDS).

The ability of an RF system to resolve spectral features can be described in terms of phase noise, defined as the ratio (in decibels) of the noise in a 1 Hz bandwidth at a specified offset f_m, to the oscillator signal amplitude at frequency f_0. In a microwave communications context, a local oscillator at frequency f_0 is used to downconvert the desired signal f_D to an intermediate frequency f_IF. We want to know what is the maximum allowable phase noise L_max (in dBc/Hz) so that the downconversion of an adjacent undesired (interference) signal f_ID is suppressed by a factor of S, called selectivity and expresssed in dB. The maximum phase noise is [84, eqn 13.50]

$$L_\mathrm{max} = D - S - I - 10\log(B),$$

where D is the desired signal level (dBm), I is the undesired (interference) signal level (dBm), and B is the bandwidth (in Hz) of the intermediate frequency.

To translate this into the language of atomic physics, the goal is no longer to convert a signal down to IF, but rather to drive a transition at a specified detuning without exciting adjacent transitions. The previously defined variables acquire new meanings:

$$\begin{aligned} f_0 &= \text{applied RF frequency} \\ f_\mathrm{D} &= \text{desired transition frequency} \\ f_\mathrm{I} &= \text{frequency of a nearby unwanted transition} \\ f_\mathrm{IF} &= \text{detuning} \end{aligned}$$

Assuming the desired (D) and undesired (I) signals are both same strength, and the signal linewidth (from molecular flythrough time) is $B \approx 10^3$ Hz, a 20 dB selectivity requires the

phase noise to be smaller than -50 dBc/Hz. This requirement is satisfied by both signal sources described below.

The DDS source is AD9910 (Analog Devices Inc., Norwood, MA), outputting out to approx. $f_{\text{DDS}} = 400$ MHz. The signal is mixed with a fixed $f_0 = 1$ GHz wave generated by one of the outputs of the SynthHD Pro (WindFreak Technologies, Holiday, FL). A low pass filter removes the sum component $f_{\text{DDS}} + f_0$ and passes the difference, to be mixed with a 12.65 GHz signal from the other SynthHD output. In the second stage, a high-pass filter has to be used, passing the sum of the frequencies, since the signal generator only goes up to 14 GHz. This also filters out a strong sub-harmonic in the high-frequency SynthHD output. (Two-stage mixing is required because of the difficulty in filtering out one of the 100 MHz sidebands from a signal exceding 10 GHz.) The resulting frequency-agile microwave signal is doubled or quadrupled if necessary, amplified, power-split with an SPDT switch, and recombined in an orthomode transducer before being led to the horn. The horns were custom engineered by Sage Millimeter (now Eravant Inc., Torrance, CA) for spot-focusing the beam energy onto the molecular beam.

F. State preparation 1

The design for the first state preparation region is currently under way.

G. Electrostatic lens

TODO: Constructed by Oskari Timgren. TBD.

H. State preparation 2

The second state preparation region has not yet been designed.

I. Interaction region

TODO: Constructed at UMass. TBD.

J. State projection and readout

To be designed.

V. PRELIMINARY MEASUREMENTS

TODO: TBD.

A. Optical cycling for detection

B. TlF beam properties

C. Rotational cooling

D. Magnetic shielding

VI. OUTLOOK

For my thesis, I expect to work to some extent on all parts of the apparatus, but especially the following:

- beam source (which I have already built myself)
- rotational cooling and other microwave systems (I have already assembled the prototypes of the systems, and partially they are already used in the preliminary testing),
- magnetic shielding (which will be mine to characterize and assemble)
- simulations of non-adiabatic transitions
- simulating rotational cooling or other aspects of the experiment, as needed.

Rough timeline to finishing thesis at end of my 6th year:

- 2020: Finish characterization of the rotational cooling. Start construction state-preparation regions. Start construction of optical cycling and detection.
- 2021: Testing of state-prep, optical cycling, detection regions. Testing interaction region, and final magnetic shielding.
- 2022: First experimental runs; taking data; data analysis; error characterisation.
- 2023: Getting to a first measurement result.

Most of the work will be carried out at the Sloane Physics Laboratory, in the spaces of Prof DeMille. Parts of the experiment have been, and will be, constructed at UMass and Columbia — for example, parts of the laser systems, and the interaction region electrodes and vacuum chamber.

REFERENCES

[1] Lev Landau. On the conservation laws for weak interactions. *Nuclear Physics*, 3(1):127–131, 1957. doi:10.1016/0029-5582(57)90061-5.

[2] EM Purcell and NF Ramsey. On the possibility of electric dipole moments for elementary particles and nuclei. *Physical Review*, 78(6):807, 1950. doi:10.1103/PhysRev.78.807.

[3] James H Christenson, Jeremiah W Cronin, Val L Fitch, and René Turlay. Evidence for the 2π decay of the K_2^0 meson. *Physical Review Letters*, 13(4):138, 1964. doi:10.1103/PhysRevLett.13.138.

[4] C. Patrignani et al. Review of Particle Physics. *Chin. Phys.*, C40 (10):100001, 2016. doi:10.1088/1674-1137/40/10/100001.

[5] R. D. Peccei. *CP* violation: A theoretical review, 1995. https://arxiv.org/abs/hep-ph/9508389.

[6] Makoto Kobayashi and Toshihide Maskawa. *CP*-violation in the renormalizable theory of weak interaction. *Progress of Theoretical Physics*, 49(2):652–657, 1973. doi:10.1143/PTP.49.652.

[7] George WS Hou. Source of *CP* violation for the baryon asymmetry of the universe. *International Journal of Modern Physics D*, 20(08):1521–1532, 2011. doi:10.1142/S0218271811019694.

[8] Maxim Pospelov and Adam Ritz. The theta term in QCD sum rules and the electric dipole moment of the vector meson. *Nuclear physics B*, 558(1-2):243–258, 1999. doi:10.1016/S0550-3213(99)00399-5.

[9] Brent Graner, Y Chen, EG Lindahl, BR Heckel, et al. Reduced limit on the permanent electric dipole moment of ^{199}Hg. *Physical review letters*, 116(16):161601, 2016. doi:10.1103/PhysRevLett.116.161601.

[10] CA Baker, DD Doyle, P Geltenbort, K Green, MGD Van der Grinten, PG Harris, P Iaydjiev, SN Ivanov, DJR May, JM Pendlebury, et al. Improved experimental limit on the electric dipole moment of the neutron. *Physical Review Letters*, 97(13):131801, 2006. doi:10.1103/PhysRevLett.97.131801.

[11] Peter W Graham, Igor G Irastorza, Steven K Lamoreaux, Axel Lindner, and Karl A van Bibber. Experimental searches for the axion and axion-like particles. *Annual Review of Nuclear and Particle Science*, 65:485–514, 2015. doi:10.1146/annurev-nucl-102014-022120.

[12] M Eh Pospelov and IB Khriplovich. Electric dipole moment of the W boson and the electron in the Kobayashi-Maskawa model. *Yadernaya Fizika*, 53(4):1030–1033, 1991.

[13] Maxim Pospelov and Adam Ritz. Electric dipole moments as probes of new physics. *Annals of physics*, 318(1):119–169, 2005. doi:10.1016/j.aop.2005.04.002.

[14] CL Bennett, D Larson, JL Weiland, N Jarosik, G Hinshaw, N Odegard, KM Smith, RS Hill, B Gold, M Halpern, et al. Nine-year Wilkinson Microwave Anisotropy Probe (WMAP) observations: final maps and results. *The Astrophysical Journal Supplement Series*, 208(2):20, 2013. doi:10.1088/0067-0049/208/2/19.

[15] Laurent Canetti, Marco Drewes, and Mikhail Shaposhnikov. Matter and antimatter in the universe. *New J. Phys. 14 (2012) 095012*, 2012. doi:10.1088/1367-2630/14/9/095012.

[16] Andrei D Sakharov. Violation of *CP* invariance, *C* asymmetry, and baryon asymmetry of the universe. *Soviet Physics Uspekhi*, 34(5):392, 1991.

[17] Antonio Riotto and Mark Trodden. Recent progress in baryogenesis. *Annual Review of Nuclear and Particle Science*, 49(1): 35–75, 1999. doi:10.1146/annurev.nucl.49.1.35.

[18] Konrad Kleinknecht and Burkhard Renk. Unitarity triangle from *CP* invariant quantities. *Physics Letters B*, 639(6):612–615, 2006. doi:10.1016/j.physletb.2006.07.015.

[19] Z. Fodor. Electroweak phase transitions. *Nuclear Physics B Proceedings Supplements*, 83–84:121–125, 2000. doi:10.1016/S0920-5632(00)91603-7Get.

[20] MS Safronova, D Budker, D DeMille, Derek F Jackson Kimball, A Derevianko, and Charles W Clark. Search for new physics with atoms and molecules. *Reviews of Modern Physics*, 90(2): 025008, 2018. doi:10.1103/RevModPhys.90.025008.

[21] LI Schiff. Measurability of nuclear electric dipole moments. *Physical Review*, 132(5):2194, 1963. doi:10.1103/PhysRev.132.2194.

[22] VV Flambaum and JSM Ginges. Nuclear Schiff moment and time-invariance violation in atoms. *Physical Review A*, 65(3): 032113, 2002. doi:10.1103/PhysRevA.65.032113.

[23] John D Jackson. *Classical Electrodynamics*. John Wiley & Sons, 1998.

[24] Harley Flanders. Differentiation under the integral sign. *The American Mathematical Monthly*, 80(6):615–627, 1973. doi:10.1080/00029890.1973.11993339.

[25] OP Sushkov, VV Flambaum, and IB Khriplovich. Possibility of investigating *P*- and *T*-odd nuclear forces in atomic and molecular experiments. *Zh. Eksp. Teor. Fiz*, 87(11):1521–1540, 1984.

[26] PV Coveney and PGH Sandars. Parity-and time-violating interactions in thallium fluoride. *Journal of Physics B: Atomic and Molecular Physics*, 16(20):3727, 1983. doi:10.1088/0022-3700/16/20/009.

[27] Donghyun Cho, K Sangster, and EA Hinds. Search for time-reversal-symmetry violation in thallium fluoride using a jet source. *Physical Review A*, 44(5):2783, 1991. doi:10.1103/PhysRevA.44.2783.

[28] VA Dzuba, VV Flambaum, JSM Ginges, and MG Kozlov. Electric dipole moments of Hg, Xe, Rn, Ra, Pu, and TlF induced by the nuclear schiff moment and limits on time-reversal violating interactions. *Physical Review A*, 66(1):012111, 2002. doi:10.1103/PhysRevA.66.012111.

[29] Kenneth S Krane, David Halliday, et al. *Introductory nuclear physics*. John Wiley & Sons, 1987.

[30] Edward A Hinds and PGH Sandars. Experiment to search for P-and T-violating interactions in the hyperfine structure of thallium fluoride. *Physical Review A*, 21(2):480, 1980. doi:10.1103/PhysRevA.21.480.

[31] Dean A Wilkening, Norman F Ramsey, and Daniel J Larson. Search for P and T violations in the hyperfine structure of thallium fluoride. *Physical Review A*, 29(2):425, 1984. doi:10.1103/PhysRevA.29.425.

[32] D Cho, K Sangster, and EA Hinds. Tenfold improvement of limits on T violation in thallium fluoride. *Physical Review Letters*, 63(23):2559, 1989. doi:10.1103/PhysRevLett.63.2559.

[33] PGH Sandars. Measurability of the proton electric dipole moment. *Physical Review Letters*, 19(24):1396, 1967. doi:10.1103/PhysRevLett.19.1396.

[34] Norman Ramsey. *Molecular beams*, volume 20. Oxford University Press, 1956.

[35] Norman F. Ramsey. A molecular beam resonance method with separated oscillating fields. *Phys. Rev.*, 78:695–699, Jun 1950. doi:10.1103/PhysRev.78.695.

[36] Donald Rodney Jr Schropp. *Search for time reversal symmetry violation in the hyperfine structure of thallium fluoride*. PhD thesis, Yale University, December 1988.

[37] A. Erdelyi, editor. *Higher transcendental functions. Volume I.*, volume I. McGraw-Hill, 1953.

[38] D Schropp Jr, Donghyun Cho, T Vold, and EA Hinds. New limits on time-reversal invariance from the hyperfine structure of thallium fluoride. *Physical review letters*, 59(9):991, 1987. doi:10.1103/PhysRevLett.59.991.

[39] Donghyun Cho. *Search for time reversal symmetry violation in thallium fluoride using a jet source*. PhD thesis, Yale University, December 1990.

[40] Nicholas R Hutzler, Hsin-I Lu, and John M Doyle. The buffer gas beam: an intense, cold, and slow source for atoms and molecules. *Chemical reviews*, 112(9):4803–4827, 2012. doi:10.1021/cr200362u.

[41] JF Barry, ES Shuman, and D DeMille. A bright, slow cryogenic molecular beam source for free radicals. *Physical Chemistry Chemical Physics*, 13(42):18936–18947, 2011. doi:10.1039/C1CP20335E.

[42] EB Norrgard, ER Edwards, DJ McCarron, MH Steinecker, D DeMille, Shah Saad Alam, SK Peck, NS Wadia, and LR Hunter. Hyperfine structure of the $B^3\Pi_1$ state and predictions of optical cycling behavior in the $X \rightarrow B$ transition of TlF. *Physical Review A*, 95(6):062506, 2017. doi:10.1103/PhysRevA.95.062506.

[43] Mikhail G Kozlov and Leonti N Labzowsky. Parity violation effects in diatomics. *Journal of Physics B: Atomic, Molecular and Optical Physics*, 28(10):1933, 1995. doi:10.1088/0953-4075/28/10/008.

[44] J.S. Coursey, D.J. Schwab, J.J. Tsai, and R.A. Dragoset. Atomic Weights and Isotopic Compositions (version 4.1), 2015. URL http://physics.nist.gov/Comp. Accessed on Apr 20, 2020.

[45] John Michael Brown, John M Brown, and Alan Carrington. *Rotational spectroscopy of diatomic molecules*. Cambridge University Press, 2003.

[46] Klaus-Peter Huber. *Molecular spectra and molecular structure: IV. Constants of diatomic molecules*. Springer Science & Business Media, 2013.

[47] HY Afeefy, JF Liebman, and SE Stein. NIST Chemistry WebBook, NIST Standard Reference Database number 69, 2018.

[48] Dmitry Budker, Derek Kimball, Derek F Kimball, and David P DeMille. *Atomic physics: an exploration through problems and solutions*. Oxford University Press, USA, 2004.

[49] Jim Napolitano J.J. Sakurai. *Modern quantum mechanics*. Addison-Wesley, 2 edition, 2011.

[50] Kurt Gottfried and Tung-Mow Yan. *Quantum mechanics: fundamentals*. Springer, 2013.

[51] James Roger Prior Angel and PGH Sandars. The hyperfine structure stark effect: I. Theory. *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences*, 305 (1480):125–138, 1968. doi:10.1098/rspa.1968.0109.

[52] B. H. Bransden and C. J. Joachain. *Physics of atoms and molecules*. Longman Scientific & Technical, 1990.

[53] LR Hunter, SK Peck, AS Greenspon, S Saad Alam, and D DeMille. Prospects for laser cooling TlF. *Physical Review A*, 85(1): 012511, 2012. doi:10.1103/PhysRevA.85.012511.

[54] Konrad Wenz. *Rotational Cooling and Coherent Effects in Molecular Beam Search for Nuclear Schiff Moment in Thallium Fluoride*. PhD thesis, Columbia University, (expected) 2021.

[55] GE Harrison, PGH Sandars, and SJ Wright. Experimental limit on the proton electric dipole moment. *Physical Review Letters*, 22(23):1263, 1969. doi:10.1103/PhysRevLett.22.1263.

[56] Harvey Gould. Search for an electric dipole moment in thallium. *Physical Review Letters*, 24(20):1091, 1970. doi:10.1103/PhysRevLett.88.071805.

[57] Eugene D Commins, Stephen B Ross, David DeMille, and BC Regan. Improved experimental limit on the electric dipole moment of the electron. *Physical Review A*, 50(4):2960, 1994. doi:10.1103/PhysRevA.50.2960.

[58] BC Regan, Eugene D Commins, Christian J Schmidt, and David DeMille. New limit on the electron electric dipole moment. *Physical review letters*, 88(7):071805, 2002. doi:10.1103/PhysRevLett.88.071805.

[59] Michael V Romalis, WC Griffith, JP Jacobs, and EN Fortson. New limit on the permanent electric dipole moment of 199 hg. *Physical Review Letters*, 86(12):2505, 2001. doi:10.1103/PhysRevLett.86.2505.

[60] JM Pendlebury, S Afach, NJ Ayres, CA Baker, G Ban, Georg Bison, Kazimierz Bodek, Martin Burghoff, Peter Geltenbort, Katie Green, et al. Revised experimental upper limit on the electric dipole moment of the neutron. *Physical Review D*, 92(9):092003, 2015. doi:10.1103/PhysRevD.92.092003.

[61] WC Griffith, MD Swallows, TH Loftus, Michael V Romalis, BR Heckel, and EN Fortson. Improved limit on the permanent electric dipole moment of ^{199}Hg. *Physical review letters*, 102(10): 101601, 2009. doi:10.1103/PhysRevLett.102.101601.

[62] VS Malinovsky and JL Krause. General theory of population transfer by adiabatic rapid passage with intense, chirped laser pulses. *The European Physical Journal D-Atomic, Molecular, Optical and Plasma Physics*, 14(2):147–155, 2001. doi:10.1007/s100530170212.

[63] BW Shore, MV Gromovyy, LP Yatsenko, and VI Romanenko. Simple mechanical analogs of rapid adiabatic passage in atomic physics. *American Journal of Physics*, 77(12):1183–1194, 2009. doi:10.1119/1.3231688.

[64] Victoria A Cho and Richard B Bernstein. Tight focusing of beams of polar polyatomic molecules via the electrostatic hexapole lens. *The Journal of Physical Chemistry*, 95(21):8129–8136, 1991. doi:10.1021/j100174a022.

[65] HG Bennewitz, W Paul, and Ch Schlier. Fokussierung polarer moleküle. *Zeitschrift für Physik*, 141(1-2):6–15, 1955. doi:10.1007/BF01327279.

[66] Robert A Berg, Lennard Wharton, William Klemperer, Alfred Büchler, and James L Stauffer. Determination of electronic symmetry by electric deflection: LiO and LaO. *The Journal of Chemical Physics*, 43(7):2416–2421, 1965. doi:10.1063/1.1697140.

[67] NJ Stone. Table of nuclear magnetic dipole and electric quadrupole moments. *Atomic Data and Nuclear Data Tables*, 90(1):75–176, 2005. doi:10.1016/j.adt.2005.04.001.

[68] William J Tango, John K Link, and Richard N Zare. Spectroscopy of K_2 using laser-induced fluorescence. *The Journal of Chemical Physics*, 49(10):4264–4268, 1968. doi:10.1063/1.1669869.

[69] Richard N Zare. My life with LIF: a personal account of developing laser-induced fluorescence. *Annual review of analytical chemistry*, 5:1–14, 2012. doi:10.1146/annurev-anchem-062011-143148.

[70] Iosif B Khriplovich and Steve K Lamoreaux. *CP violation without strangeness: electric dipole moments of particles, atoms, and molecules*. Springer, 2012.

[71] Zack Lasner. *Order-of-magnitude-tighter bound on the electron electric dipole moment*. PhD thesis, Yale University, May 2019.

[72] S Millman. On the determination of the signs of nuclear magnetic moments by the molecular beam method of magnetic resonance. *Physical Review*, 55(7):628, 1939. doi:10.1103/PhysRev.55.628.

[73] C Alden Mead. The 'noncrossing' rule for electronic potential energy surfaces: The role of time-reversal invariance. *The Journal of Chemical Physics*, 70(5):2276–2283, 1979. doi:10.1063/1.437733.

[74] Thomas Dittrich, Peter Hänggi, Gert-Ludwig Ingold, Bernhard Kramer, Gerd Schön, and Wilhelm Zwerger. *Quantum transport and dissipation*, volume 3. Wiley-Vch Weinheim, 1998.

[75] Jon H Shirley. Solution of the Schrödinger equation with a Hamiltonian periodic in time. *Physical Review*, 138(4B):B979, 1965. doi:10.1103/PhysRev.138.B979.

[76] H. J. Metcalf and P. van der Straten. *Laser cooling and trapping*. Springer-Verlang, 2002.

[77] The HDF Group. Hierarchical Data Format version 5, Mar 2018.

[78] Moreton Moore. Symmetrical intersections of right circular cylinders. *The Mathematical Gazette*, 58(405):181–185, 1974. doi:10.2307/3615957.

[79] DJ McCarron, SA King, and SL Cornish. Modulation transfer spectroscopy in atomic rubidium. *Measurement science and technology*, 19(10):105601, 2008. doi:10.1088/0957-0233/19/10/105601.

[80] Gary C Bjorklund. Frequency-modulation spectroscopy: a new method for measuring weak absorptions and dispersions. *Optics letters*, 5(1):15–17, 1980. doi:10.1364/OL.5.000015.

[81] Fei Zi, Xuejian Wu, Weicheng Zhong, Richard H Parker, Chenghui Yu, Simon Budker, Xuanhui Lu, and Holger Müller. Laser frequency stabilization by combining modulation transfer and frequency modulation spectroscopy. *Applied optics*, 56(10): 2649–2652, 2017. doi:10.1364/AO.56.002649.

[82] Daniel A Steck. Cesium D line data, 2003.

[83] RWP Drever, John L Hall, FV Kowalski, J Hough, GM Ford, AJ Munley, and H Ward. Laser phase and frequency stabilization using an optical resonator. *Applied Physics B*, 31(2):97–105, 1983. doi:10.1007/BF00702605.

[84] David M Pozar. *Microwave Engineering 3e*. Wiley, 2006.